



Genetic Improvement: Taking real-world source code and improving it using computational search methods

Alexander Brownlee,



Markus Wagner,



John R. Woodward



Latest version of slides at https://sandybrownlee.github.io/files/GI_tutorial_PPSN_2026.pdf

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<https://doi.org/10.1145/3795101.3807748>

Instructors

UNIVERSITY of
STIRLING



- Alexander (Sandy) Brownlee is an Associate Professor in the Division of Computing Science and Mathematics at the University of Stirling. His main topics of interest are in search-based optimisation methods and machine learning, with applications in civil engineering, transportation and SBSE. Within SBSE, he is interested in automated bug-fixing and improvement of non-functional properties such as run-time and energy consumption; how these different objectives interact with each other; and novel approaches to mutating code. He is also one of the developers of Gin, an open-source toolkit for experimentation with Genetic Improvement on real-world software projects.
- Markus Wagner is an Associate Professor at the Department of Data Science and AI, Monash University, Australia. His research includes mathematical runtime analysis of heuristic optimization algorithms, theory-guided algorithm design, and applications of heuristic methods to software engineering and renewable energy production. He has led industry-funded projects by Google, Facebook, and other companies in defense, mining, and energy. He has authored about 200 articles and has attracted over AUD 10M in funding. His awards include one best poster, one best presentation, four best papers, one medal, and one Humies Gold Award, and the 2024 German IT Security Award.



Instructors

- John R. Woodward is Head of School of Maths and Computer Science at Heriot Watt University. He has also been a lecturer at the University of Stirling and the University of Nottingham. John holds a BSc in Theoretical Physics, an MSc in Cognitive Science, and a PhD in Computer Science from the University of Birmingham. His research interests include Automated Software Engineering, AI/Machine Learning, and Genetic Programming. He has over 50 publications and has given more than 50 talks at international conferences. John's experience spans industrial, military, educational, and academic settings, including employment with EDS, CERN, RAF, and three UK universities.
- <https://scholar.google.com/citations?user=ypG5tTsAAAAJ&hl=en&inst=16061989973938494330>



Overview

● Introduction: why GI? And basic principles	25 min	[John]
● Recent research questions	15 min	[Markus]
● The human perspective	15 min	[John]
● Noteworthy papers & connections to other topics	15 min	[Markus]
● Demonstration: Gin and Magpie	30 min	[Sandy]
● Open Challenges and Q&A	10 min	[Markus]

computer improves it



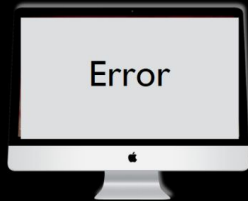
Justyna Petke

Functional Properties

LOGICAL



New Feature



Bug Repair

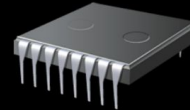
accuracy

Non-Functional Properties

PHYSICAL



Execution Time



Memory



Bandwidth



Battery



Size

UNITS

There is nothing correct about a flat battery
(BILL LANGDON)

What is Genetic Improvement

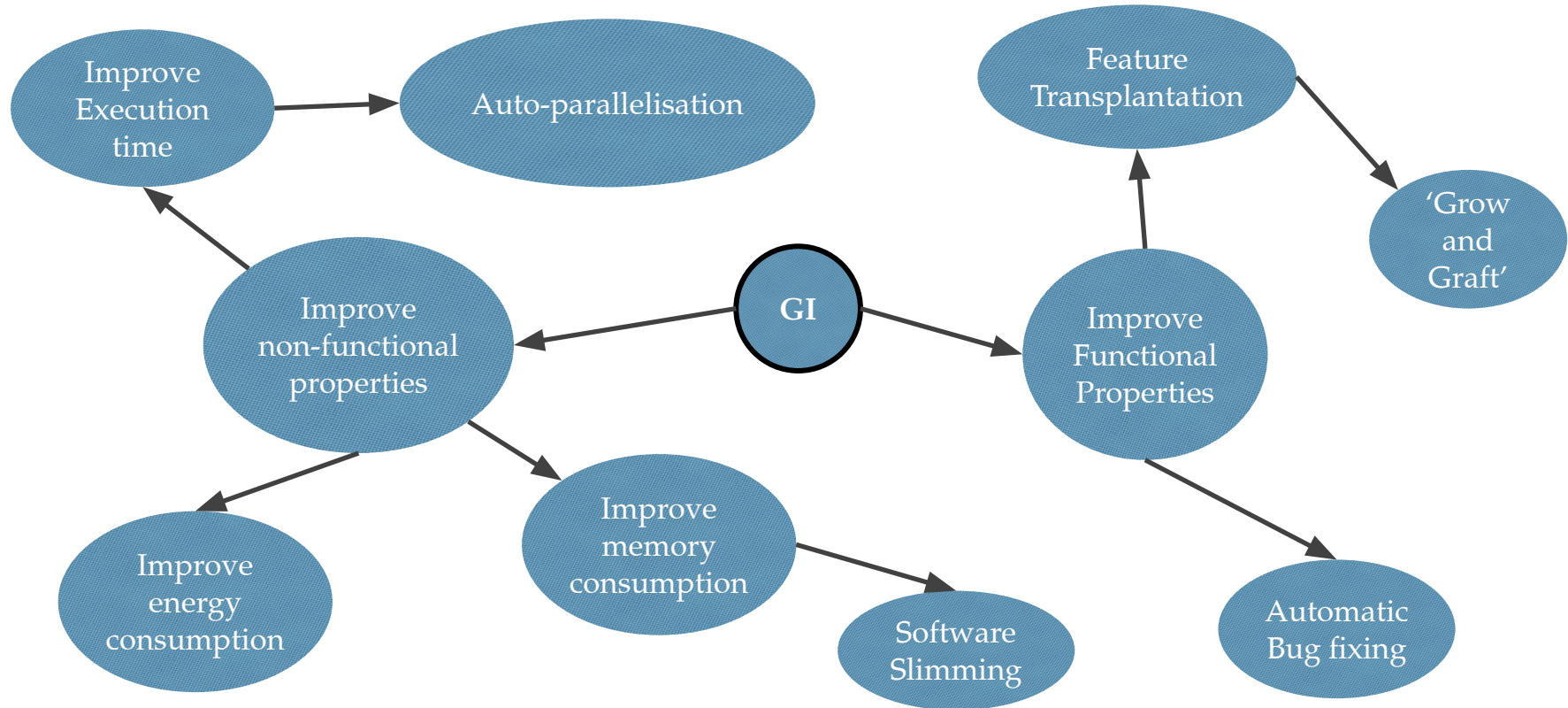
A wordy definition:

Genetic Improvement is the application of search-based (typically evolutionary) techniques to modify software with respect to some user-defined fitness measure.

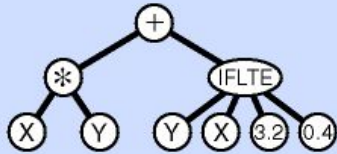


It's just GP - BUT starting
with a **nearly complete**
program
[Wolfgang Banzhaf]

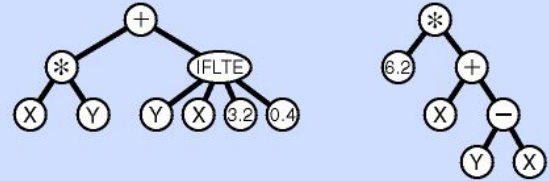
What is Genetic Improvement



Genetic Programming overview



mutation



crossover

Genetic Programming: GI's ROOTS

1. **Aim** – *to discover new programs by telling the computer what we want it to do, but not how we want it to do it* – John Koza
2. **How** – we evolve computer programs using natural selection.
3. **Starts from scratch (empty program)**
4. Choose **primitives** (terminal set/FEATURES and function set)
5. Choose **representation** (tree based, graph based, linear e.g. CGP)
6. Choose *fitness function, parameters, genetic operators.*

GI forces “the full capabilities of programming languages”- side effects, ADFs, LOOPS

GP vs GI: if you can't beat them, join them.

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ABSTRACT

Genetic Programming (GP) has been criticized for targeting irrelevant problems [12], and is true of the wider machine

(procedures, methods, macros, routines), and so GI has to deal with the reality of existing software systems. However, most of the GP literature is not concerned with Tur-

Popular Science

On the next few slides: a few easy to digest articles for non-specialists.

<https://theconversation.com/computers-will-soon-be-able-to-fix-themselves-are-it-departments-for-the-chop-85632>

Computers will soon be able to fix themselves – are IT departments for the chop?

October 12, 2017 3.29pm BST

IT?



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<https://theconversation.com/how-computers-are-learning-to-make-human-software-work-more-efficiently-43798>

How computers are learning to make human software work more efficiently

June 25, 2015 10.08am BST



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<https://theconversation.com/never-mind-the-iphone-x-battery-life-could-soon-take-a-great-leap-forward-83901>

THE CONVERSATION

Academic rigour, journalistic flair

🔍 Search analysis, research, academics...

Arts + Culture Business + Economy Cities Education Environment + Energy Health + Medicine Politics + Society **Science + Technology** Brexit

Never mind the iPhone X, battery life could soon take a great leap forward

September 13, 2017 2.29pm BST



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Jerry Swan

Competent Programmers Hypothesis

1. programmers write programs that are almost perfect.
2. program faults are syntactically small (slip of finger, T/F)
3. corrected with a few keystrokes. (e.g. < for <=)
4. **GI can find small patches.**
5. Small changes are non-unique (write 7 lines code, or utter 7 words **before they're unique**)

Plastic Surgery Hypothesis.

the content of new code can often be assembled out of fragments of code that already exist.

Barr et al. [71] showed that changes are 43% graftable from the exact version of the software being changed.

The Plastic Surgery Hypothesis: Changes to a codebase contain snippets that already exist in the codebase at the time of the change, and these snippets can be efficiently found and exploited.

THE CODE CONTAINS SOLUTIONS – CANDIDATE PATCHES

Representations of PROGRAMS

Natural Representation of CODE

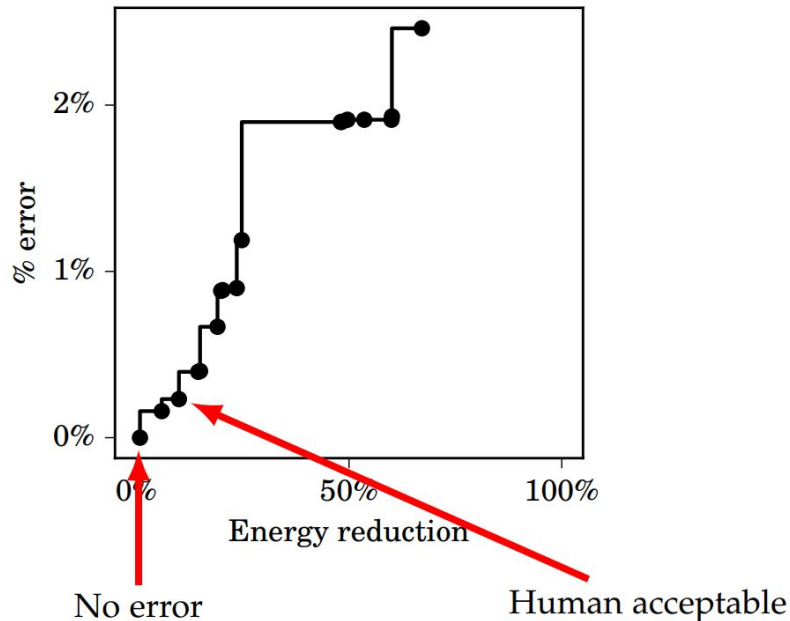
1. Text files e.g. Program.java is a text file.
2. Abstract syntax tree (AST) – Genprog, Genofix.
3. Java byte code (also C binaries) [102]
4. Errors, compile, halting (Langdon - discard)

Objectives

- Functional (**logical properties**)
 - Accuracy e.g. as in machine learning. FLOAT
 - Number of bugs – as measured against a set of test cases. BOOLEAN/INTEGER
 - New functionality – e.g. ...
- Non-functional (*physical* **properties**)
 - Execution time
 - Energy (power consumption – peak/average)
 - Memory
 - Bandwidth
- Multi-objective
 - a set of trade-off programs = a single tuneable program

Multi-Objective

- Seems be convex
- – simple argument (see pic)
- Can provide a set of programs
- weighted sum of objectives?
- weight has meaning to user.
- *Will there be elbow/knee points?*



Slow connections....



Loading Gmail



Loading standard view | [Load basic HTML](#) (for slow connections)

GISMOE

The GISMOE challenge:

to create an automated program development environment in which the Pareto program surface is automatically constructed to support dialog with and decision making by the software designer concerning the trade offs present in the solution space of programs for a specific programming problem.

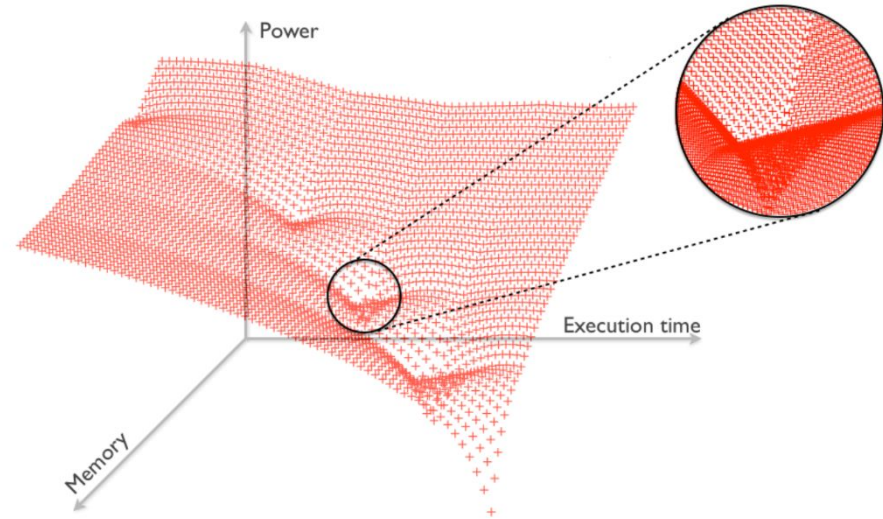


Figure 1: The GISMOE Pareto Program Surface

EDIT Operators – changes to programs

- Line level
 - Single Character level
 - Function/module level.
 - AST – GIN, Gen-0-fix, genprog,
 - Java – machine code – java byte code.
-
- LIST OF EDITS IS A PATCH.

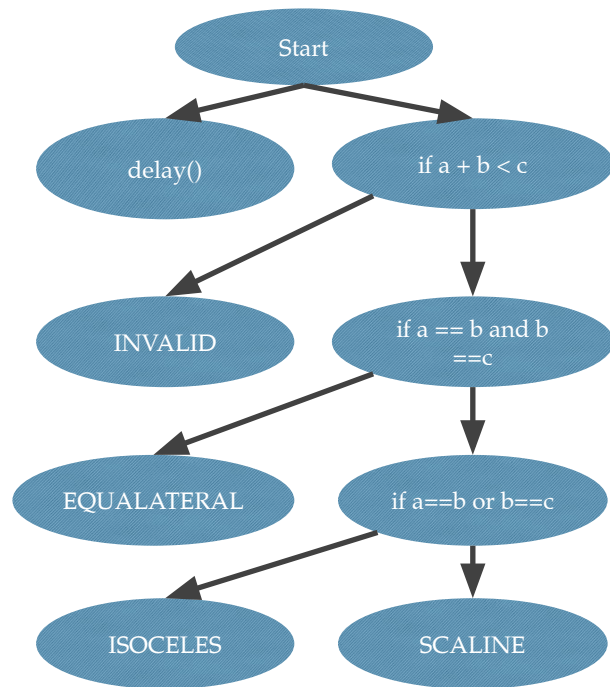
GI: An example of execution time optimisation

```
static final int INVALID = 0;
static final int SCALENE = 1;
static final int EQUALATERAL = 2;
static final int ISOCELES = 3;

public static int classifyTriangle(int a, int b, int c) {
    delay();

    assert(a <= b && b <= c);
    if (a + b <= c) {
        return INVALID;
    } else if (a == b && b == c) {
        return EQUALATERAL;
    } else if (a == b || b == c) {
        return ISOCELES;
    } else {
        return SCALENE;
    }
}

private static void delay() {
    try {
        Thread.sleep(100);
    } catch (InterruptedException e) {
        // do nothing
    }
}
```

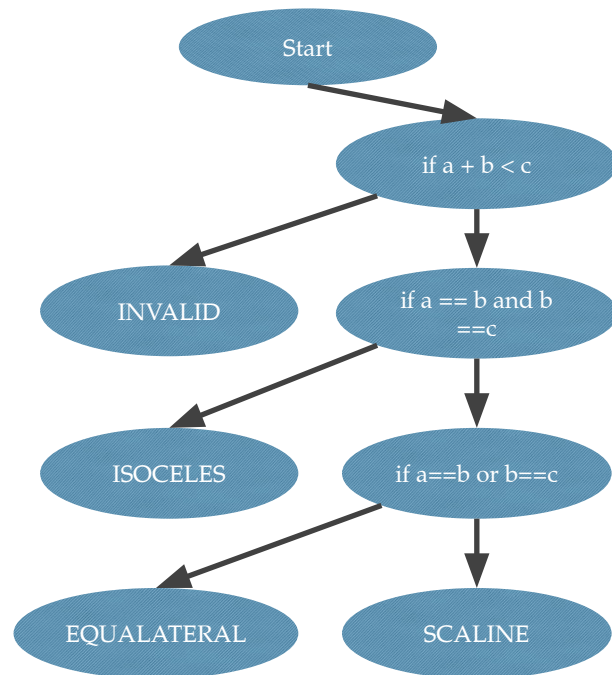


GI: An example of automated bug fixing

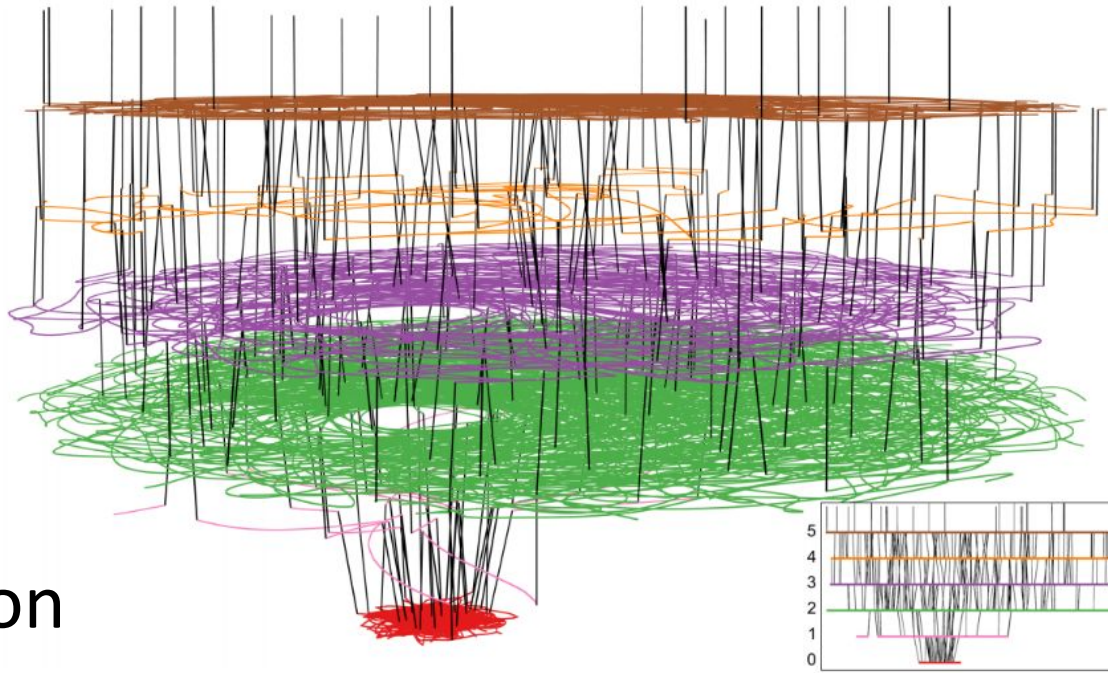
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}
```



Neutral
networks
Graceful
degradation

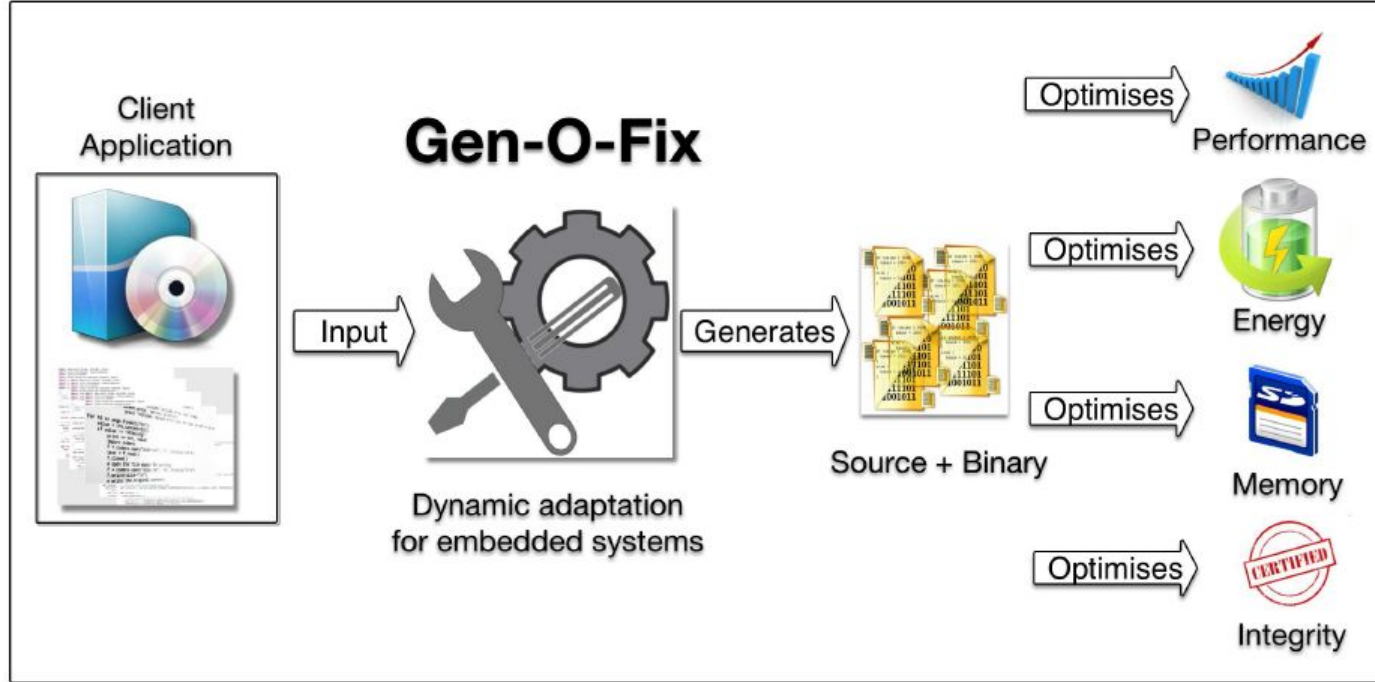


structure

Hill
climber

Fig. 1. Local optima network of the Triangle Program using 100 random starts (see Section 4.4). Edges are coloured if they start and end at the same fitness. Insert shows fitness levels edge on. Best (bottom) red 0 (pass all tests), pink 1 (fail only one test), green 2, purple 3, orange 4, brown 5.

System Diagram for Gen-O-Fix



Gen-O-Fix: Abstract Syntax Trees

Main features of framework are

1. **Embedded** adaptively.
2. Minimal end-user requirements.
 1. Initial source code: **location** of Scala source code file containing a function
 2. Fitness function: providing a means of **evaluating the quality** of system
3. **Source to source transformations**
4. Operates on **ASTs** (i.e. arbitrarily fine).

AST - scala

Code as data, data as code.

```
// code to data:
```

```
var m = 2; var x = 3; var c = 4  
val expr = reify( ( m * x ) + c )  
println( "AST = " + showRaw( expr.tree ) )
```

```
// output:
```

```
AST = Apply( Select( Apply( Select( Select( Ident("m") ,  
"elem" ) , " $times" ) , List( Select( Ident("x" ) ,  
"elem" ) ) ) , " $plus" ) , List( Select( Ident("c" ) , "elem" ) ) ) )
```



```
// run AST datatype as code:  
println( "eval = " + expr.tree.eval() )
```

```
// output:  
eval = 10
```

Gen-O-Fix Reactive Stocks

Darth-Vader



valves are simulated

... Gen-O-Fix Empire ...

⚙️ Gem-O-Fix Polynomial Stock Predictor

John Woodward (Stirling)

```
*: Some(((x: Double) => 0.4515267279707613*(x)*(x)*(x)+(0.15731708770383146*(  
    x)*x).unary_+$plus)+(0.2599003224041494*(x)))/(0.529762)))  
*: Some(((x: Double) => 0.4515267279707613*(x)*(x)*(x)+(0.15731708770383146*(  
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```

GI HashCode tuning

1. **Hadoop** provides a mapReduce implementation in Java.
2. Equals method has to obey **contract** (Reflective, Symmetric, Transitive, ...)
3. `x.equals(y)` **implies** `hashCode(x) == hashCode(y)`.
4. `hashCode` method is an integer function of a subset of an object's fields

Some GP Settings

1. Terminal set is

1. Field values
2. Random integers [0, 100]

2. Function set is

1. {+, *, XOR, AND}

3. Fitness function: close to uniform distribution of hashes (uniform distribution is the ideal), over 10,000 instances.

Distribution of Hashcodes

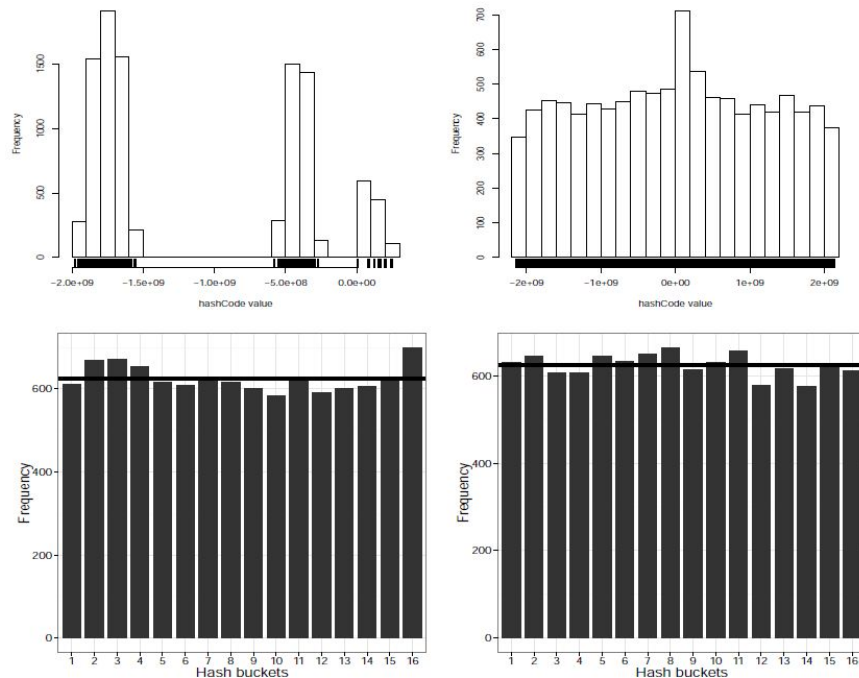


Fig. 1: The distribution of the hashcode values (top) and the distribution of the created objects in hash buckets (bottom), generated by the Apache commons (left) and the evolved function (right)

Overview

- | | | |
|---|--------|----------|
| ● Introduction: why GI? And basic principles | 25 min | [John] |
| ● Recent research questions | 15 min | [Markus] |
| ● The human perspective | 15 min | [John] |
| ● Noteworthy papers & connections to other topics | 15 min | [Markus] |
| ● Demonstration: Gin and Magpie | 30 min | [Sandy] |
| ● Open Challenges and Q&A | 10 min | [Markus] |

Theory

- Hard!
- NFL not really valid for GP, and therefore GI.
 - Why – because many programs share same functionality... and the NFL would assume that all program are equally likely (which is not the case in practical applications)

=> GI will remain empirical for years to come

Theory

E.g. Landscapes...
Lots of neutrality!

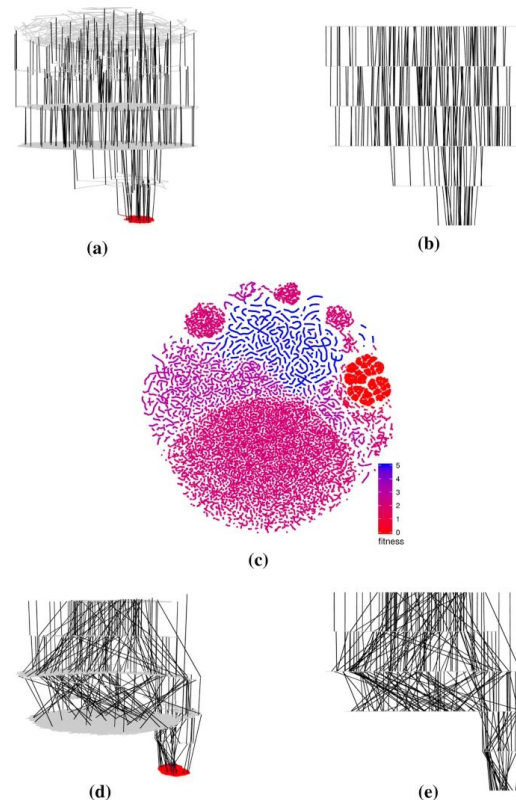
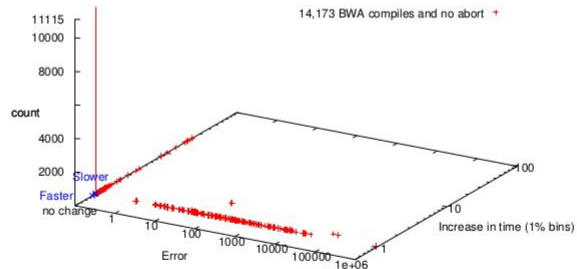
Veerapen N, Ochoa G. Visualising the global structure of search landscapes: genetic improvement as a case study. Genetic programming and evolvable machines. 2018. 19(3):317-49

Software is Not Fragile

William B. Langdon and Justyna Petke

CREST Department of Computer Science,
University College London Gower Street, London WC1E 6BT, UK

Abstract. Trying all simple changes (first order mutations) to executed C, C++ and CUDA source code shows software engineering artefacts are more robust than is often assumed. Of those that compile, up to 89% run without error. Indeed a few mutants are improvements. Program fitness landscapes are smoother. Analysis of these programs, a parallel nVidia GPGPU kernel, all CUDA samples and the GNU C library shows many lines of code and integer values are repeated and may follow Zipf's law.



Theory

E.g. Sampling of the space

Genetic Programming and Evolvable Machines (2019) 20:531–580
<https://doi.org/10.1007/s10710-019-09355-3>



A journey among Java neutral program variants

Nicolas Harrand¹ · Simon Allier² · Marcelino Rodriguez-Cancio³ ·
Martin Monperrus¹ · Benoit Baudry¹

Received: 18 December 2018 / Revi:
© The Author(s) 2019

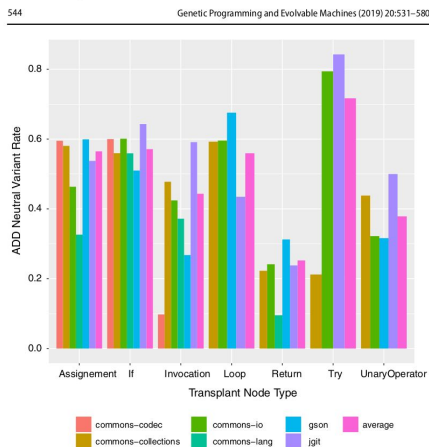


Fig. 6 Neutral variant rate for the app transformation, depending on the type of AST node used as trans-

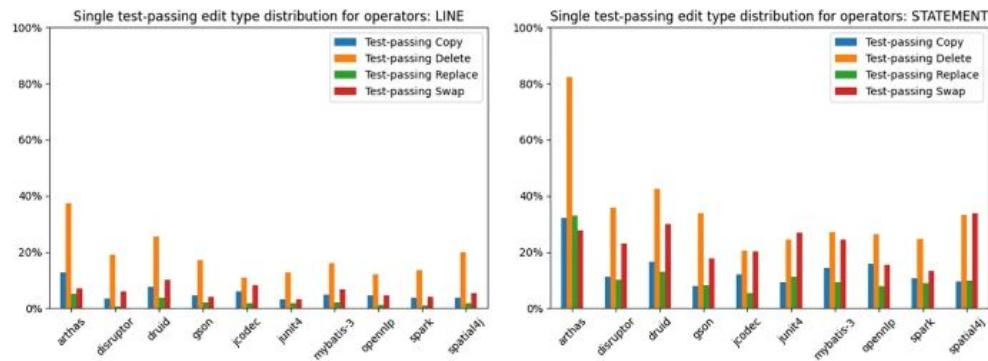
Empirical Software Engineering (2023) 28:104
<https://doi.org/10.1007/s10664-023-10344-5>



Program transformation landscapes for automated program modification using Gin

Justyna Petke¹ · Brad Alexander² · Earl T. Barr¹ · Alexander E.I. Brownlee³ ·
Markus Wagner⁴ · David R. White⁵

Accepted: 23 May 2023 / Published online: 14 July 2023
© The Author(s) 2023



Theory

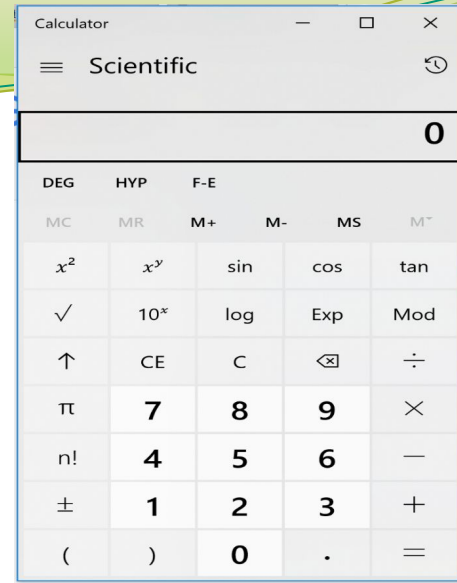
Lots remains!

Where and when does GI work best?

How does this vary for functionality / run time / energy ... ?

GI & Benchmarking

1. GP suffered a “midlife crisis”
2. Toy problem e.g. lawnmower
3. Genetic Programming Needs Better Benchmarks [White et al.]
4. Machine Learning that Matter [Wagstaff 2012] what is 1% meaning
5. *Is Software Engineering the best benchmark for GP?*
6. Do we have a stable set of benchmarks for GI?
(for program repair: <http://program-repair.org/benchmarks.html>)
7. Blot, Aymeric, and Justyna Petke. "A Comprehensive Survey of Benchmarks for Automated Improvement of Software's Non-Functional Properties" arXiv:2212.08540 (2022).
8. Benchmarking is more complex (noise, hardware, prog lang, ...)



Measuring Energy

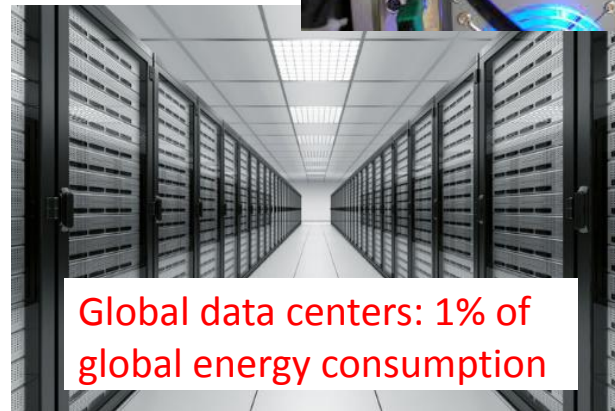
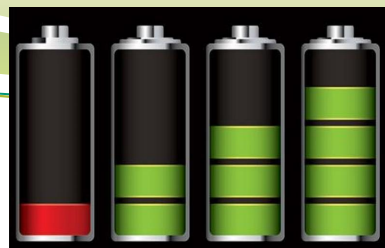
- computational energy consumption growing importance, particularly at the extremes (i.e., mobile devices and datacentres).

one line = one unit

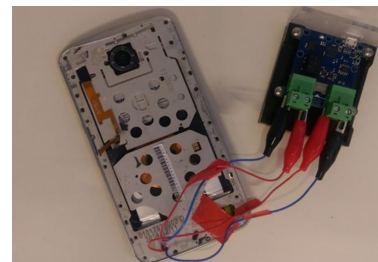
simulate (runtime/system calls/) Tools Opacitor, PowerGauge

read battery indicator

physically measure and validate (e.g. see Bokhari et al.)



Global data centers: 1% of global energy consumption



GI@GECCO'17 Deep Parameter Optimisation on Android Smartphones for Energy Minimisation - A Tale of Woe and a Proof-of-Concept

CEC 2019 Mind the gap - a distributed framework for enabling energy optimisation on modern smart-phones in the presence of noise, drift, and statistical insignificance [#19776]

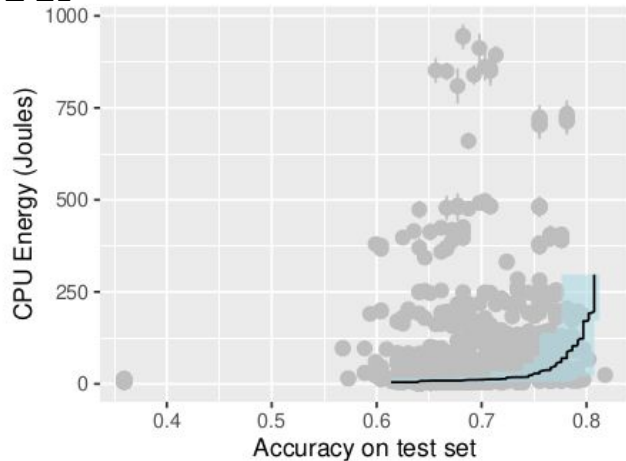
Measuring Energy

Trade-offs to exploit, but lots of noise and many confounding factors

Exploring the Accuracy – Energy Trade-off in
Machine Learning

Alexander E.I Brownlee, Jason Adair, Saemundur O. Haraldsson and John Jabbo

GI@ICSE'21



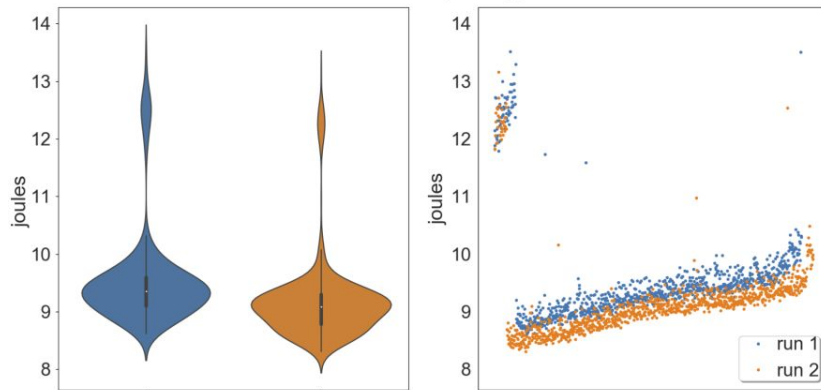
**Towards Rigorous Validation of Energy Optimisation
Experiments**

Mahmoud A. Bokhari

Brad Alexander, Markus Wagner

GECCO '20

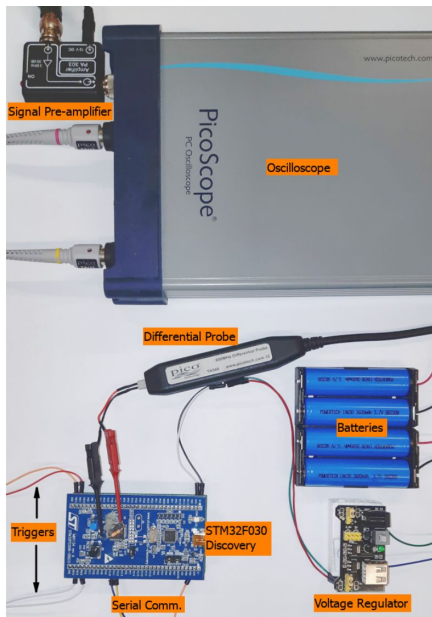
Rebound Library Energy Use



Shown:
measurements
of the same
task... just the
phone was
recharged and
restarted
between run1
and run2

GI to eliminate side-channels

Minimise the “signal” when performing cryptographic operations, as they can leak information on the secret key via power consumption profiling!



CCS'22

ROSITA: Towards Automatic Elimination of Power-Analysis Leakage in Ciphers

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NDSS'21

ROSITA++: Automatic Higher-Order Leakage Elimination from Cryptographic Code

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GI to improve Speed

Also noisy... also platform dependent (read: you can specialise code to architectures)



PLDI'23 Distinguished Paper
Humies'23 Gold Award
2024 German IT Security Award



Key aspects:

- Demonstrated on straight line code (cryptographic primitives)
- Optimisation at the levels of: intermediate representation & Assembly
- *Automatic formal proofs of correctness*
- Yields new, fastest implementations
- Curve25519 Code in BoringSSL

⇒ this runs in your Chrome and Edge browser now, i.e. on billions of devices around the world!



CryptOpt: Verified Compilation with Randomized Program Search for Cryptographic Primitives

JOEL KUEPPER, University of Adelaide, Australia
ANDRES ERBSEN, Massachusetts Institute of Technology, USA
JASON GROSS, Massachusetts Institute of Technology, USA
OWEN CONOLY, Massachusetts Institute of Technology, USA
CHUYUE SUN, Stanford University, USA
SAMUEL TIAN, Massachusetts Institute of Technology, USA
DAVID WU, University of Adelaide, Australia
ADAM CHLIPALA, Massachusetts Institute of Technology, USA
CHITCHANOK CHUENGSAITANSUP, The University of Melbourne, Australia
DANIEL GENKIN, Georgia Institute of Technology, USA
MARKUS WAGNER, Monash University, Australia
YUVAL YAROM, Ruhr University Bochum, Germany

<https://github.com/0xADE1A1DE/CryptOpt>

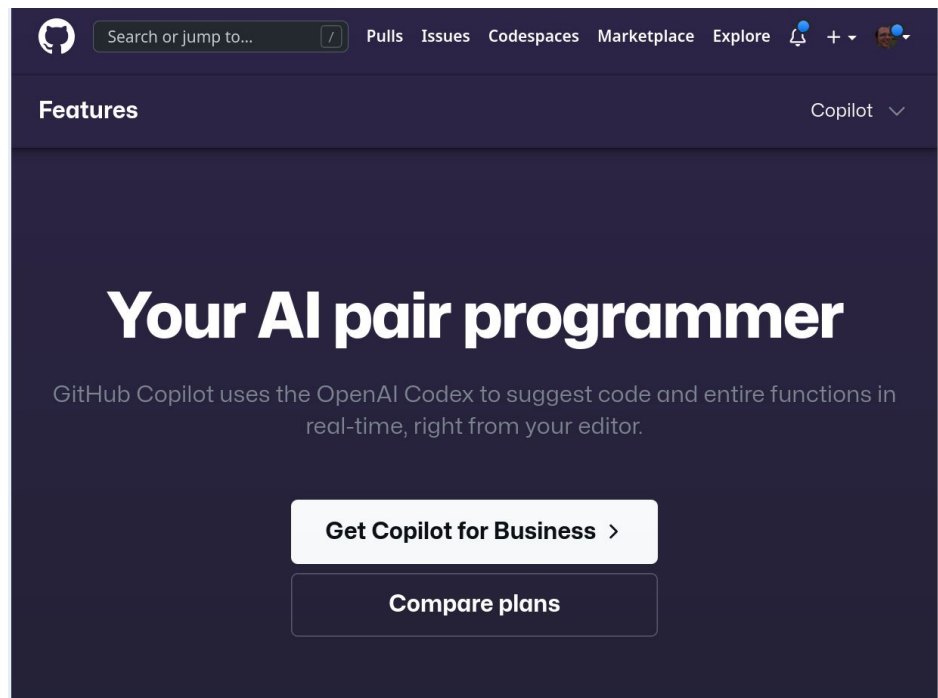
What about Copilot/ChatGPT...?

Large language models generate code!

Replicate patterns given some prompt

Can lead to errors!* Related-but-incorrect solutions

AI search tests the code as it goes, so can be constrained to only produce variants that (probably) work



*Jones E & Steinhardt J. Capturing failures of large language models via human cognitive biases.
In AH Oh, A Agarwal, D Belgrave & K Cho, eds., Advances in Neural Information Processing Systems. 2022

Towards Objective-Tailored Genetic Improvement Through Large Language Models

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12th International Workshop on Genetic Improvement
ICSE 2023

Enhancing Genetic Improvement Mutations Using Large Language Models

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Callan²[0000-0002-5692-6203], Karine Even-Mendoza³[0000-0002-3099-1189], Alina
Geiger⁴[0009-0002-3413-283X], Carol Hanna²[0009-0009-7386-1622], Justyna
Petke²[0000-0002-7833-6044], Federica Sarro²[0000-0002-9146-442X], and
Dominik Sobania⁴[0000-0001-8873-7143]

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SSBSE Challenge Track 2023

An Analysis of the Automatic Bug Fixing Performance of ChatGPT

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University College London
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2023 IEEE/ACM International Workshop on
Automated Program Repair (APR), 2023

Creative and Correct: Requesting Diverse Code Solutions from AI Foundation Models

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Singapore, Singapore

FORGE 2024

Overview

- | | | |
|---|--------|----------|
| ● Introduction: why GI? And basic principles | 25 min | [John] |
| ● Recent research questions | 15 min | [Markus] |
| ● The human perspective | 15 min | [John] |
| ● Noteworthy papers & connections to other topics | 15 min | [Markus] |
| ● Demonstration: Gin and Magpie | 30 min | [Sandy] |
| ● Open Challenges and Q&A | 10 min | [Markus] |

The human perspective

E. R. Winter et al., "Let's Talk With Developers, Not About Developers: A Review of Automatic Program Repair Research," in *IEEE Transactions on Software Engineering*, doi:

<https://doi.org/10.1109/TSE.2022.3152089>

V. Nowack et al., "Expanding Fix Patterns to Enable Automatic Program Repair," 2021 IEEE 32nd International Symposium on Software Reliability Engineering (ISSRE), 2021, pp. 12-23, doi:

<https://10.1109/ISSRE52982.2021.00015> .

E. Winter et al., "How do Developers Really Feel About Bug Fixing? Directions for Automatic Program Repair," in *IEEE Transactions on Software Engineering*, vol. 49, no. 4, pp. 1823-1841, 1 April 2023, doi:

[10.1109/TSE.2022.3194188](https://doi.org/10.1109/TSE.2022.3194188).

Kuhail, Mohammad Amin, et al. "“Will I be replaced?” Assessing ChatGPT's effect on software development and programmer perceptions of AI tools." *Science of Computer Programming* 235 (2024): 103111.

<https://doi.org/10.1016/j.scico.2024.103111>

Eladawy, Hadeel, Claire Le Goues, and Yuriy Brun.

"Automated program repair, what is it good for? not absolutely nothing!." *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*. 2024.

Meem, Fairuz Nawar, Justin Smith, and Brittany Johnson.

<https://doi.org/10.1145/3597503.3639095>

"Exploring experiences with automated program repair in practice." *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*. 2024.

<https://doi.org/10.1145/3597503.3639182>

How do Developers Really Feel About Bug Fixing? Directions for Automatic Program Repair

1. Bug fixing is not universally disliked
2. Bug fixing remains largely manual and individual
3. Developers are cautiously optimistic about Automatic Program Repair (APR)
4. Strong preference for “developer-in-the-loop” APR
5. Trust and correctness matter more than speed
6. Attitudes toward APR are widely shared across demographics

"“Will I be replaced?” Assessing ChatGPT's effect on software development and programmer perceptions of AI tools.”

- Two empirical studies assessed ChatGPT's role in software development:
 - (1) code-generation performance on 180 LeetCode problems, and
 - (2) programmer perceptions via a survey of 99 professionals.
- ChatGPT 3.5 performs well on easy and medium problems but struggles significantly with hard problems; success rate drops sharply as problem complexity increases.
- Problem popularity shows a weak positive effect on ChatGPT's success rate, suggesting better performance on widely discussed or represented problems.
- Programmers primarily use AI tools for boilerplate code, code explanation, research, debugging, and documentation, reporting moderate productivity gains.
- Job security concerns are modest today but higher for the future.



Let's Talk With Developers, Not About Developers: A Review of Automatic Program Repair Research

1. Developers are frequently cited but rarely studied
2. Claims about developer needs are often unsubstantiated
3. Existing APR human studies are few and of mixed quality
4. Most human studies are tool-specific and hard to generalize
5. Professional developers are under-represented
6. The paper calls for a developer-centred research agenda



Automated Program Repair, What Is It Good For? Not Absolutely Nothing!

1. Large controlled user study.
2. APR suggestions strongly influence outcomes.
3. Correct APR suggestions are extremely valuable.
4. Deceptive APR suggestions are harmful.
5. Experience level does not mitigate risk.
6. Overall perception is positive.



Towards Developer-Centered Automatic Program Repair: Findings from Bloomberg

- 1.The paper presents one of the first in-depth industrial studies of APR use
- 2.APR development at Bloomberg shifted from a technical to a user-centered mindset,
- 3.Developers view APR fixes as pragmatic “starting points,” not final answers
- 4.Workflow fit is critical for acceptance,
- 5.Trust is built incrementally
- 6.The authors recommend a developer-in-the-loop future for APR

Exploring experiences with automated program repair in practice

1.331 software practitioner study

2.Awareness $\neq$ adoption

3.Who is most aware: Developers, small companies, >5 years' experience

4.APR used less than alternatives: human support, GitHub, Stack Overflow, and ChatGPT.

5.Key barriers to adoption.

6.Design implications.

Overview

- | | | |
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| ● Open Challenges and Q&A | 10 min | [Markus] |

Case study

Fixing bugs: A real world example of GI in action!

S.O. Haraldsson, John R. Woodward, Alexander E. I. Brownlee, and Kristin Siggeirsdottir. 2017. Fixing bugs in your sleep: how genetic improvement became an overnight success. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '17). ACM, New York, NY, USA, 1513-1520. DOI: <https://doi.org/10.1145/3067695.3082517>

S. O. Haraldsson, J. R. Woodward and A. I. E. Brownlee, "The Use of Automatic Test Data Generation for Genetic Improvement in a Live System," 2017 IEEE/ACM 10th International Workshop on Search-Based Software Testing (SBST), Buenos Aires, 2017, pp. 28-31. DOI: <https://10.1109/SBST.2017.10>

S.O. Haraldsson, 2017. 'Genetic Improvement of Software: From Program Landscapes to the Automatic Improvement of a Live System', PhD thesis, University of Stirling, Stirling. <http://hdl.handle.net/1893/26007>

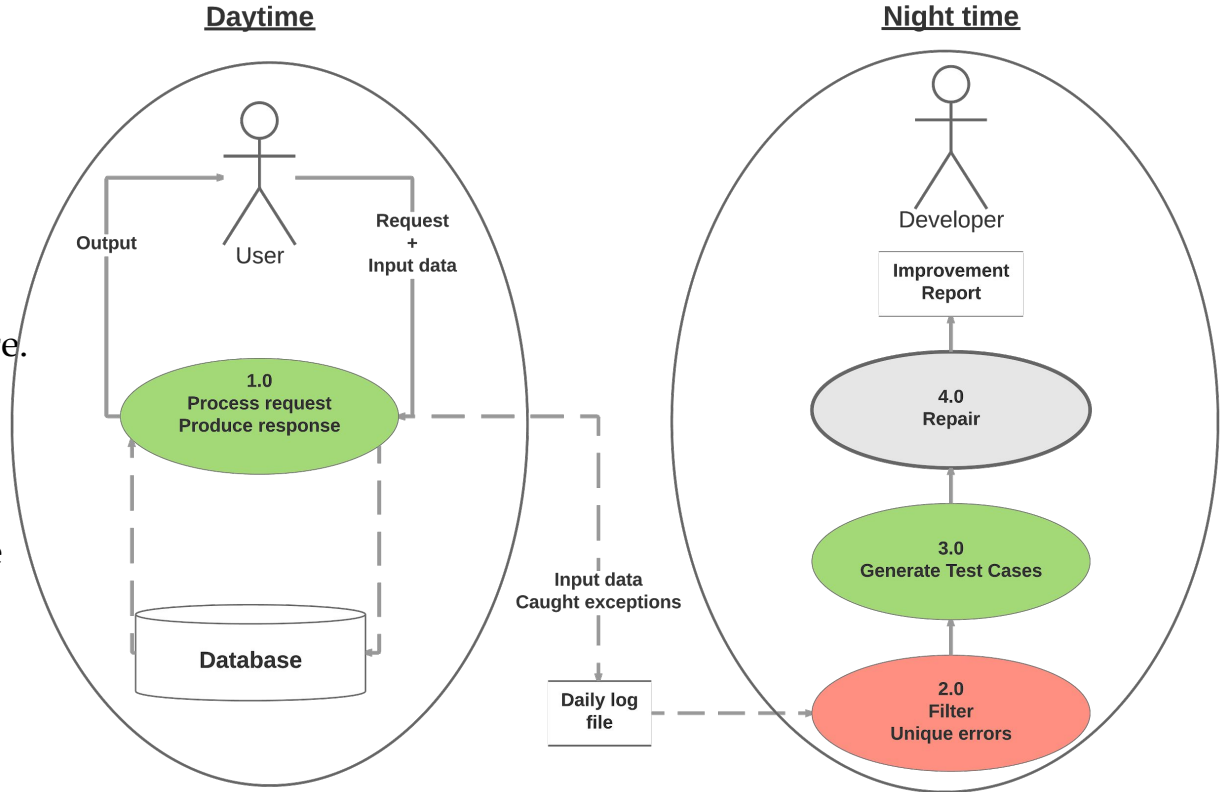
S.O. Haraldsson, John R. Woodward, Alexander E. I. Brownlee, Albert V. Smith, and Vilmundur Gudnason. 2017. Genetic improvement of runtime and its fitness landscape in a bioinformatics application. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '17). ACM, New York, NY, USA, 1521-1528. DOI: <https://doi.org/10.1145/3067695.3082526>

S.O. Haraldsson, 2017. 'Genetic Improvement of Software: From Program Landscapes to the Automatic Improvement of a Live System', PhD thesis, University of Stirling, Stirling. <http://hdl.handle.net/1893/26007>

G. B. Saemundsdottir, and S.O. Haraldsson 2024. "Large Language Models as All-in-One Operators for Genetic Improvement.", to appear in Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24). ACM, New York, NY, USA, 1501-1508. URL: <https://doi.org/10.1145/3638530.3654408>

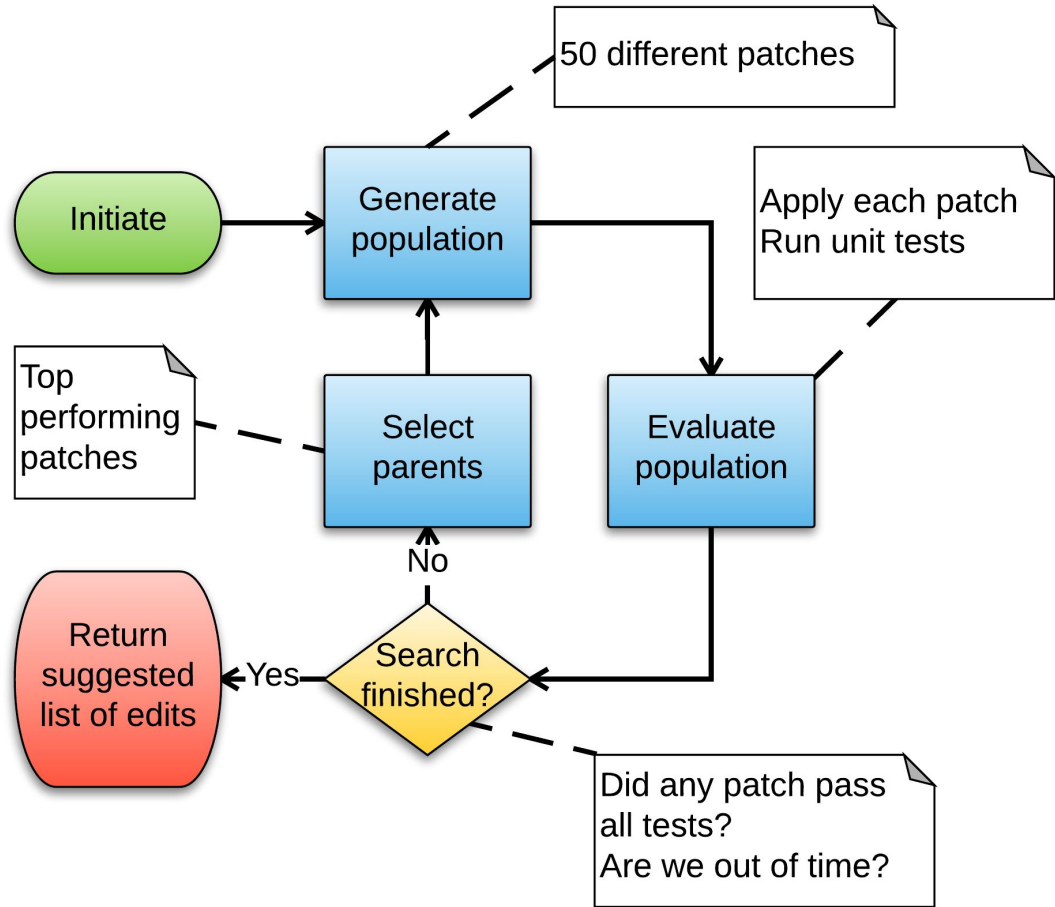
When last user logs out

1. Procedure 2.0 started
 - Sorts and filters the day's exceptions
2. Procedure 3.0
 - Emulates input data, type, size and structure.
 - Produces test cases
3. Procedure 4.0
 - Genetic Improvement
 - Parallel process on the server
 - Outputs report for developer



- Procedure 4.0

- Genetic Improvement
 - Pop.= 50 patches
 - fit.= #passed tests
 - select= $\frac{1}{2}$ pop by fitness
 - Output= report



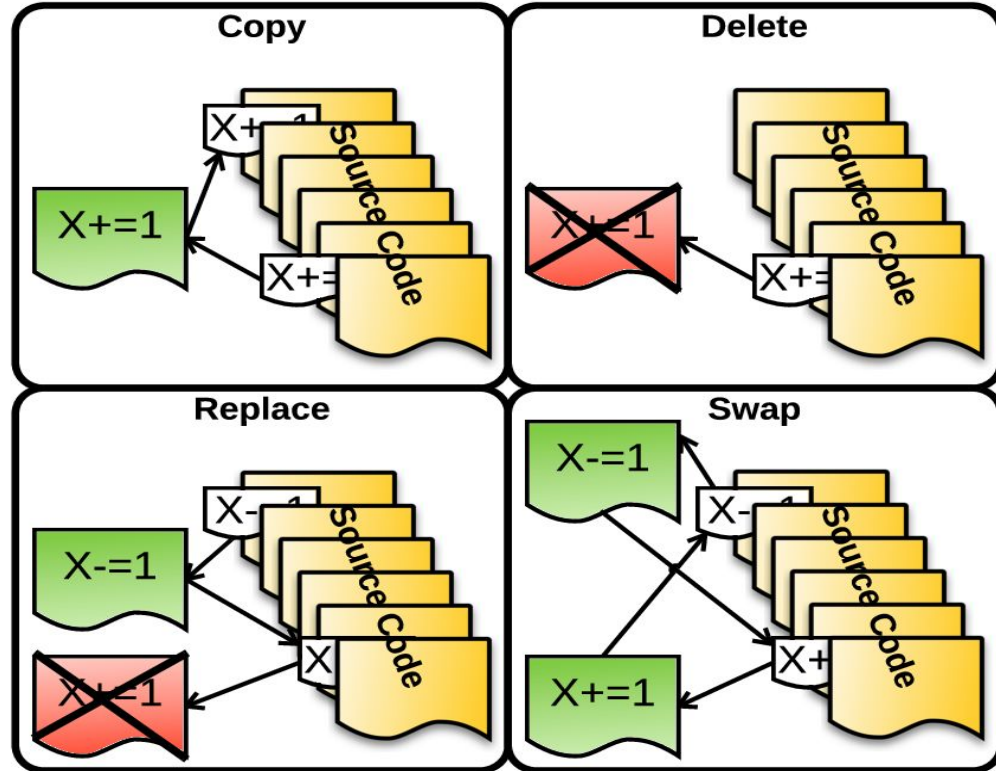
4 different types of implemented **Edits**

Primitive types:

- **Copy**
 - Equivalent to:
CTRL+C -> CTRL+V
- **Delete**
 - Almost what you think

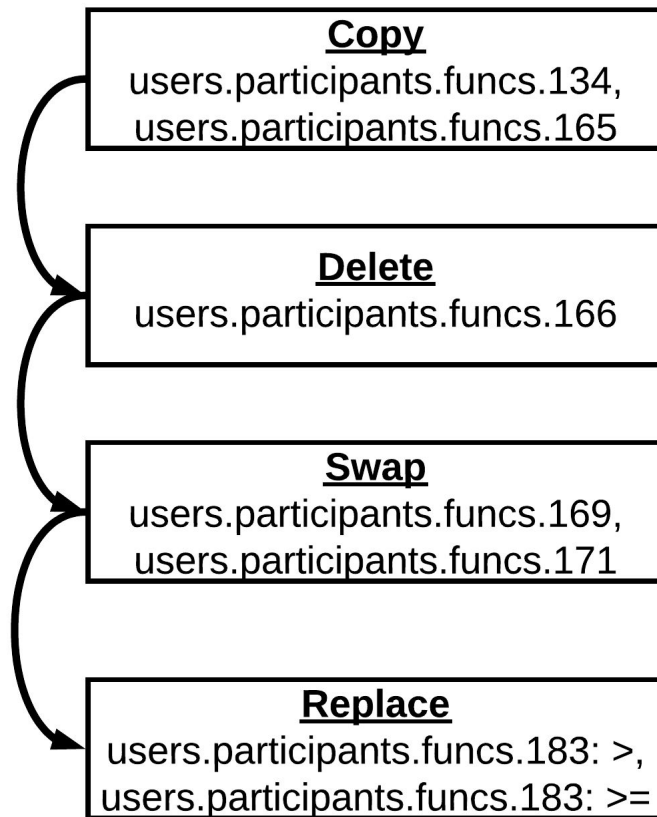
Composite types:

- **Replace**
 - Copy + Delete
- **Swap**
 - 2x Copy + 2x Delete



A list of edits makes a suggestion

- Reads like a recipe
 - Step-by-step
- Automatically reduced
 - Delta debugging
- Scrutinised by the developer
 - Might change the recipe



Or just let an LLM do the work

Points to keep in mind though:

Representation

Of the Code

Of the Changes

Readability

To the human

Reliability

Of the Prompts

Of the User

Efficiency vs Workload distribution

Prompt engineering

vs

Computations (The GI search)

vs

The LLM Model

Improving CUDA DNA Analysis Software with Genetic Programming (2015)

W.B. Langdon , B.Y.H. Lam , J. Petke & M. Harman



A 50,000 line
system

1. DNA sequencing
2. consisting of 8,000+ lines of code.
3. improved version is up to 3x faster
4. downloaded 1,000 times.
5. **Ported by IBM** to one of their super computers

Optimising Existing Software with Genetic Programming

William B. Langdon and Mark Harman

- Bowtie2, a DNA sequence alignment/sequence analysis tool
- Using Genetic Improvement, Harman and Langdon were capable of increasing performance 70x.

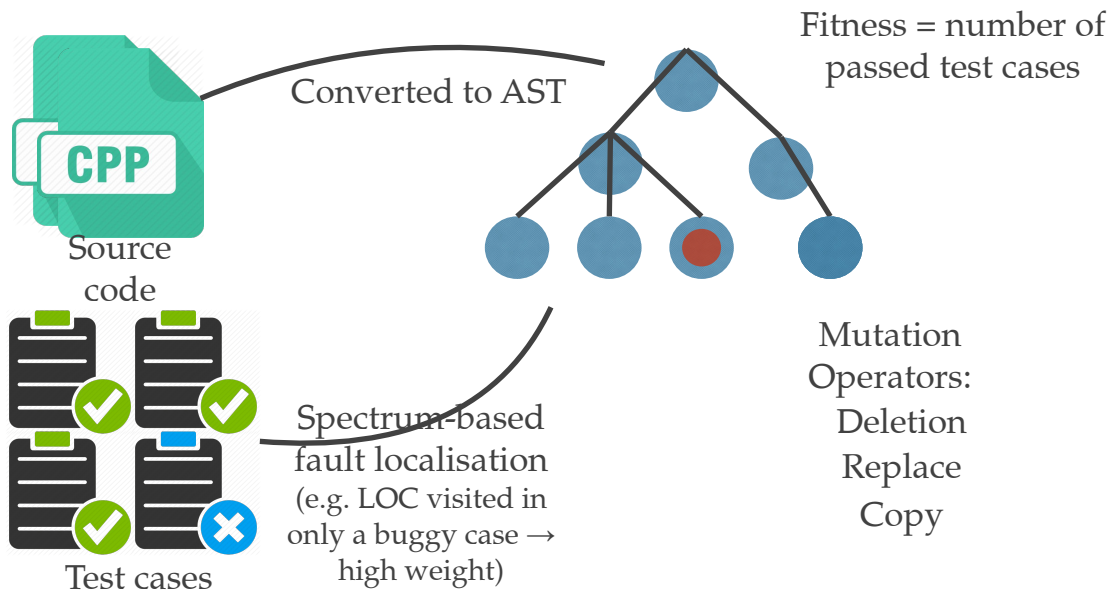
A Systematic Study of Automated Program Repair: Fixing 55 out of 105 Bugs for \$8 Each

(2012)
Cited >700 times

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- Where an adequate test suite is provided, GenProg has been shown to fix **real-world bugs**
- It has **inspired a variety of alternative frameworks**, most of which claim to outperform GenProg

Automated Software Transplantation

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CREST, University College London, Malet Place, London, WC1E 6BT, UK
{e.barr,m.harman,yue.jia,alexandru.marginean.13,j.petke}@ucl.ac.uk



Donor



Host



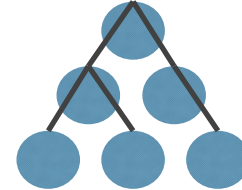
(2015)

Babel Pidgin: SBSE Can Grow and Graft Entirely New Functionality into a Real World System

Mark Harman, Yue Jia, and William B. Langdon

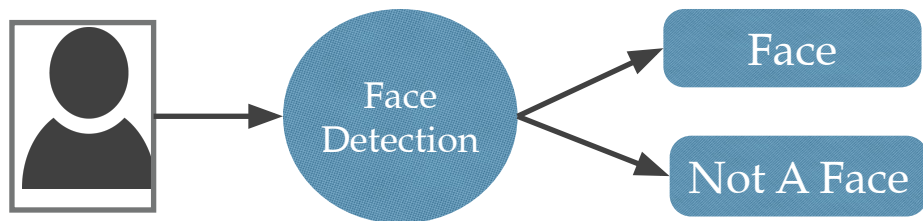
University College London, CREST centre, UK

English to Korean;
English to Portuguese



Deep Parameter Optimisation for Face Detection Using the Viola-Jones Algorithm in OpenCV

Bobby R. Bruce^{1(✉)}, Jonathan M. Aitken^{2(✉)}, and Justyna Petke^{1(✉)}



Integer
Literals
extracted

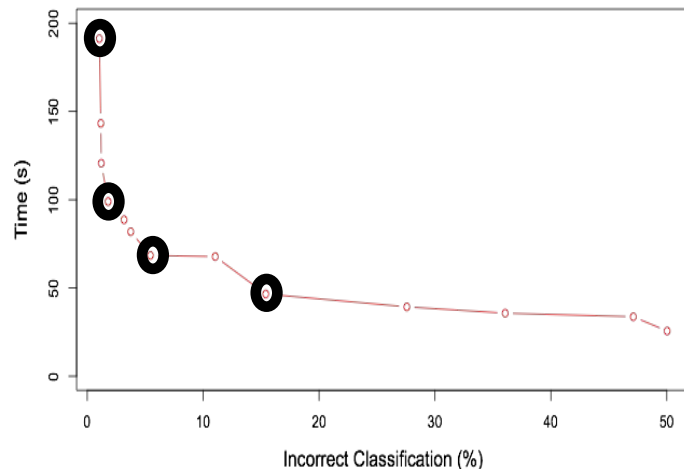
Int Literal	Value
line_1_a	1
line_1_b	22
line_10_a	5
line_12_a	2

To a genotype



Multi-objective
optimisation

Original: 191s, 1.04% inaccuracy
99s (48% decrease), 1.8% inaccuracy
68s (64% decrease), 5.4% inaccuracy
46s (76% decrease), 15.4% inaccuracy



Genetic Improvement of Data gives double precision invsqrt

W. B. Langdon

Department of Computer Science, University College London Gower Street, WC1E 6BT, UK
w.langdon@cs.ucl.ac.uk

ABSTRACT

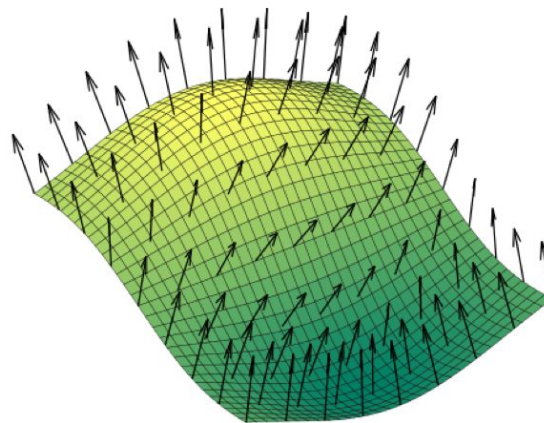
CMA-ES plus manual code changes rapidly transforms 512 Newton-Raphson start points from a GNU C library table driven version of sqrt into a double precision reciprocal square root function. The GI $x^{-\frac{1}{2}}$ is far more accurate than Quake's InvSqrt, Quare root.

CCS CONCEPTS

• **Software and its engineering** → **Search-based software engineering**;

KEYWORDS

evolutionary computing, Evolution Strategies, GGGP, search based software engineering, SBSE, software maintenance of literals, data transplantation, glibc, Newton's method



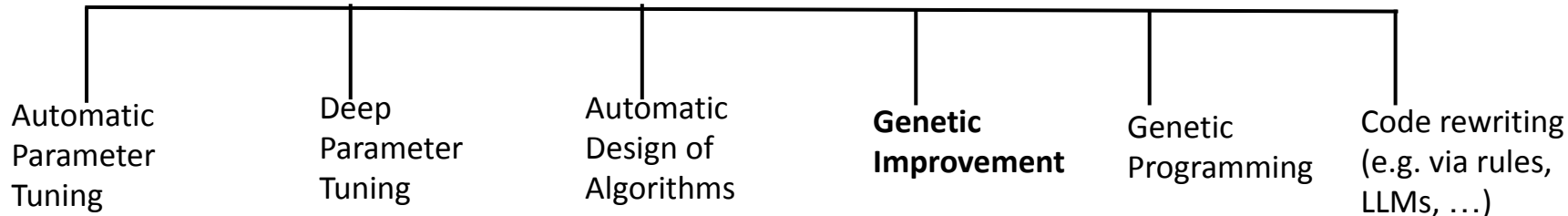
Phd Theses

- David R. White
- Andrea Arcuri
- Bobby R. Bruce
- Sæmundur Ó. Haraldsson
- Mahmoud R. Bokhari
- Michail Basios

...<https://geneticimprovementofsoftware.com/learn/survey> lists 33 to date

Relationship to other fields

- Optimization/machine learning - OVERFITTING (or: specialisation?)
("Is the cure worse than the disease?" Smith et al. FSE 2015)
- Genetic Programming and Metaheuristics
- the automatic design of algorithms
- Automatic parameter tuning/deep parameter tuning/GI



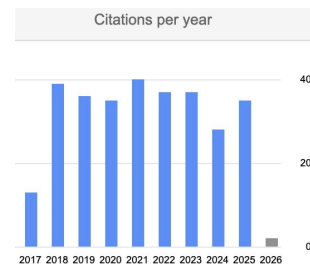
Starting point – Surveys

IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION

Genetic Improvement of Software: a Comprehensive Survey

Justyna Petke, Saemundur O. Haraldsson, Mark Harman,
William B. Langdon, David R. White, and John R. Woodward

IEEE TEVC 2017



GI@GECCO'19

A Survey of Genetic Improvement Search Spaces

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ACM Computing Surveys 2025

A Comprehensive Survey of Benchmarks for Improvement of Software's Non-Functional Properties

AYMERIC BLOT, Université de Rennes, Rennes, France, Université du Littoral Côte d'Opale, Calais, France,
and University College London, London, United Kingdom of Great Britain and Northern Ireland
JUSTYNA PETKE, University College London, London, United Kingdom of Great Britain and Northern
Ireland

Starting point – Surveys

2023 IEEE/ACM International Conference on Software Engineering: Future of Software Engineering (ICSE-FoSE)

Large Language Models for Software Engineering: Survey and Open Problems

This survey is not
about GI, but GI is
prominently
featured

Angela Fan
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Beliz Gokkaya
PyTorch Team
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Mark Harman
Instagram Product Foundation
Meta Platforms Inc.
London, UK

Mitya Lyubarskiy
Developer Infrastructure
Meta Platforms Inc.
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Shubho Sengupta
FAIR
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Menlo Park, CA, USA

Shin Yoo
School of Computing
KAIST
Daejeon, Korea

Jie M. Zhang
Department of Informatics
King's College London
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If you want to stay up to date, check out this *living survey* on GI

<https://geneticimprovementofsoftware.com/learn/survey>

Most recent GI Workshop



The **15th** International Workshop on Genetic Improvement @ICSE 2026 (13 April)

- Keynote from Dr. Wesley Klewerton Guez Assunção on “GI for Software Modernization, Synergy and Opportunities”
- Papers:
 - Applying Genetic Improvement Techniques for Automated Program Repair of Transpiled Code
 - Databending as a Target for Genetic Improvement
 - GI-Agent Search-Based LLM Agent for Code Optimization with Genetic Improvement
 - Improving a Parallel C++ Intel SSE SIMD Linear Genetic Programming Interpreter
- Future workshops <http://geneticimprovementofsoftware.com>

Overview

- | | | |
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| ● Open Challenges and Q&A | 10 min | [Markus] |

GI toolkits

GI:

Gin, PyGGI, MAGPIE, locoGP

C, C++, and Java

APR: ASTOR, GenProg

See also: Zuo, Blot, Petke: *Evaluation of genetic improvement tools for improvement of non-functional properties of software*. GECCO '22



Demonstration: Gin

Gin

Genetic Improvement in No time
Toolbox for GI research targeting Java

<https://github.com/gintool/gin>

Initiated by David White, developed through community effort



Current active development is on LLM branch:

<https://github.com/gintool/gin/tree/llm>

Gin's Goals / Principals

A toolkit to enable experimentation

Remove *incidental* difficulties of GI for research and teaching

Work on open-source software projects out-of-the-box

Designed to be hacked and extended

Gin's Goals / Principals

Maven™

 **Gradle** Build Tool

JAVAPARSER

FOR PROCESSING JAVA CODE




Apache Commons™
<http://commons.apache.org/>



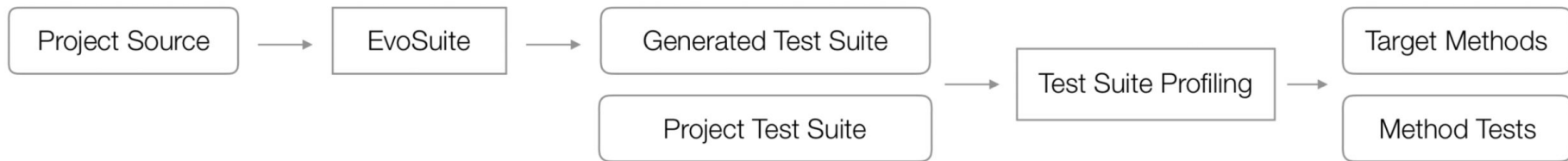
Java

Libraries

License

Gin Pipelines

Preprocessing



Search Space Analysis



Could be random, or search for improvement

Edits and Searches

- Edits are single changes to source code; multiple edits = 1 patch
- Gin supports various edit types:
 - line level (Langdon) - delete/replace/copy/swap/move
 - statement level (GenProg) - delete/replace/copy/swap/move
 - micro (OR $\Leftrightarrow$ AND) ($++ \Leftrightarrow --$); reorder expressions ($X \ \&\& \ Y \Leftrightarrow Y \ \&\& \ X$)...
 - LLM edits based on templates (local ollama, and OpenAI API)

- Also built-in searches:

- EmptyPatchTester
- RandomSampler
- DeleteEnumerator
- LocalSearch
- GP (single objective population-based)
- NSGA-II

```
2 public class Test {
3     public int strangeAdd(int a, int b) {
4         if (a == b) {
5             return 1000;
6         } else {
7             return a + b;
8         }
9     }
10 }
```

← line ↓ statement

```
2 public class Test {
3     public int strangeAdd(int a, int b) {
4         if (a == b) {
5             return 1000;
6         } else {
7             return a + b;
8         }
9     }
10 }
```

Edits and Searches

- Gin also provides tools to make designing your own edits easier so that you can focus on higher-level tasks, for example:
 - “Tell me which lines are eligible for deletion in this method”
 - “Delete this line”
 - “Give me all the for loop conditions in this method”
- Similarly, extend the “Sampler” class to implement your own searches
 - Utility methods provided to run and analyse unit tests against the patched code

Example edit

```
1 public class ReplaceStatement extends StatementEdit {
2
3     public int sourceID;
4     public int destinationID;
5
6     public ReplaceStatement(SourceFileTree sf, Random r) {
7         sourceID = sf.getRandomStatementID(false, r);
8         destinationID = sf.getRandomStatementID(true, r);
9     }
10
11     public SourceFile apply(SourceFileTree sf) {
12         Statement source = sf.getStatement(sourceID);
13         Statement dest = sf.getStatement(destinationID);
14         return sf.replaceNode(dest, source.clone());
15     }
16
17 }
```

Disclaimer: this is a simplified version to illustrate. Some detail (e.g. serialisation, and code to avoid replacing statements within the same parent node) is omitted

Patch / Edit Evaluation

Gin invokes test cases via Junit and tracks:

- compile success;
- run-time errors, exception types
- actual & expected outcomes
- timing: wall-clock and CPU time; peak memory; (energy coming soon)

```
UnitTest[] ut = {
    new UnitTest("TriangleTest", "testInvalidTriangles"),
    new UnitTest("TriangleTest", "testEqualateralTriangles"),
    new UnitTest("TriangleTest", "testIsocelesTriangles"),
    new UnitTest("TriangleTest", "testScaleneTriangles")
};

UnitTest.defaultTimeoutMS = 10000;
int reps = 1;

SourceFileTree sf = new SourceFileTree("examples/triangle/Triangle.java",
    Collections.singletonList("classifyTriangle(int,int,int)"));

InternalTestRunner tr = new InternalTestRunner("TriangleTest",
    "examples/triangle", Arrays.asList(ut));

// Start with the empty patch
Patch patch = new Patch(sf);

// Run empty patch and log
UnitTestResultSet rs = tr.runTests(patch, reps);

boolean compiled = rs.getCleanCompile();
boolean test0TimedOut = rs.getResults().get(0).getTimedOut();
long test0ExecutionTime = rs.getResults().get(0).getExecutionTime();
String test0ExceptionMessage = rs.getResults().get(0).getExceptionMessage();
```

Build tool integration

- Maven and Gradle API documentation is sparse!
 - And many projects seem to break conventions about paths, resources etc.
- Project class wraps most of what we have learned
 - determine the project's classpath
 - find a particular source file in project file hierarchy
 - provide a standard method signature for a given method
 - provide a list of project tests
 - run a unit test given its name
- Gin can infer the necessary classpath and dependencies for running unit tests from a Maven or Gradle project, or these can be specified manually
- Maven projects can be updated automatically with new unit tests from *EvoSuite*

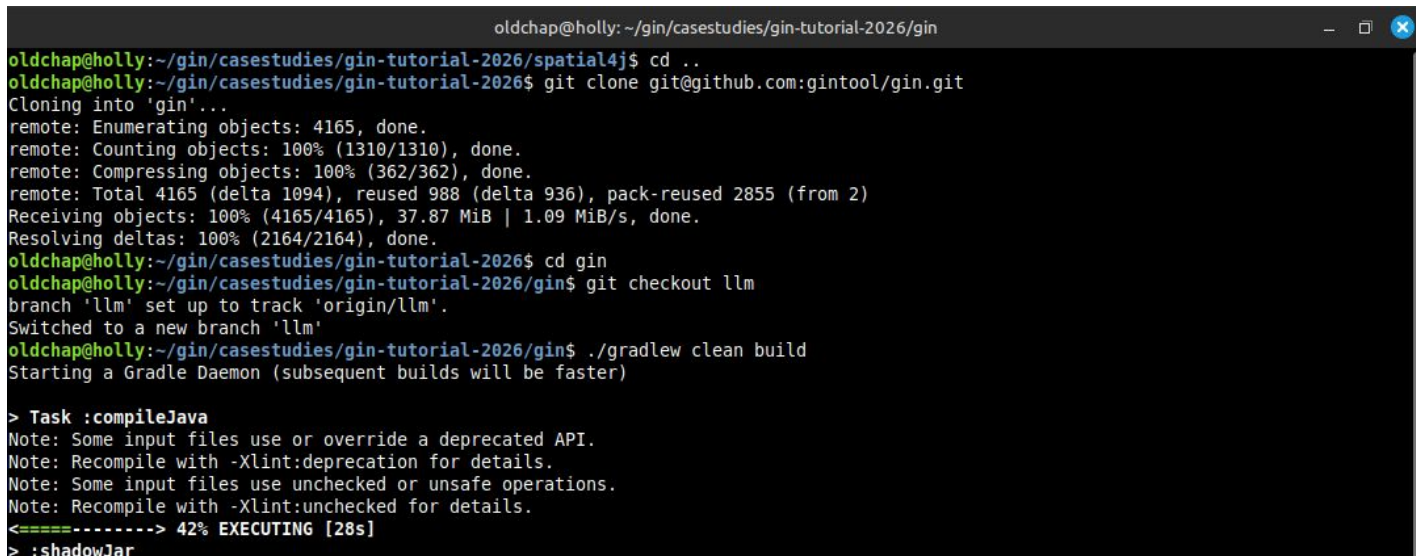
Worked example

Goal: test a new LLM-based edit for improving runtime in Spatial4J

1. Clone and build gin
2. Clone and build target project
3. Profiling
4. Test the Empty Patch
5. Implement prompt template
6. Random search
7. Local search (runtime)
8. Results

Clone and build gin

```
git clone git@github.com:gintool/gin.git
cd gin
git checkout llm
./gradlew clean build
```

A terminal window titled 'oldchap@holly: ~/gin/casestudies/gin-tutorial-2026/gin' showing the execution of the commands from the previous block. The output shows the cloning process, switching to the 'llm' branch, and the start of a Gradle build. The build task is ':compileJava' and it shows some deprecation warnings. The progress bar indicates 42% completion.

```
oldchap@holly: ~/gin/casestudies/gin-tutorial-2026/gin
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ cd ..
oldchap@holly:~/gin/casestudies/gin-tutorial-2026$ git clone git@github.com:gintool/gin.git
Cloning into 'gin'...
remote: Enumerating objects: 4165, done.
remote: Counting objects: 100% (1310/1310), done.
remote: Compressing objects: 100% (362/362), done.
remote: Total 4165 (delta 1094), reused 988 (delta 936), pack-reused 2855 (from 2)
Receiving objects: 100% (4165/4165), 37.87 MiB | 1.09 MiB/s, done.
Resolving deltas: 100% (2164/2164), done.
oldchap@holly:~/gin/casestudies/gin-tutorial-2026$ cd gin
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/gin$ git checkout llm
branch 'llm' set up to track 'origin/llm'.
Switched to a new branch 'llm'
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/gin$ ./gradlew clean build
Starting a Gradle Daemon (subsequent builds will be faster)

> Task :compileJava
Note: Some input files use or override a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
Note: Some input files use unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
<=====> 42% EXECUTING [28s]
> :shadowJar
```

Clone project

```
cd ..
```

```
git clone git@github.com:locationtech/spatial4j.git
```

```
cd spatial4j
```

```
ls
```

```
oldchap@holly:~/gin/casestudies/gin-tutorial-2026$ git clone git@github.com:locationtech/spatial4j.git
Cloning into 'spatial4j'...
remote: Enumerating objects: 24899, done.
remote: Counting objects: 100% (102/102), done.
remote: Compressing objects: 100% (62/62), done.
remote: Total 24899 (delta 33), reused 73 (delta 19), pack-reused 24797 (from 1)
Receiving objects: 100% (24899/24899), 13.63 MiB | 1.04 MiB/s, done.
Resolving deltas: 100% (8324/8324), done.
oldchap@holly:~/gin/casestudies/gin-tutorial-2026$ cd spatial4j/
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ ls
about.md  asl-v20.txt  CHANGES.md  CONTRIBUTING.md  devnotes.md  eclipse  FORMATS.md  notice.md  pom.xml  README.md  src
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$
```

Build

mvn clean compile

mvn clean test

```
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.159 s - in org.locationtech.spatial4j.shape.TestShapes2D
[INFO] Running org.locationtech.spatial4j.shape.impl.BBoxCalculatorTest
[INFO] Tests run: 100, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.075 s - in org.locationtech.spatial4j.shape.impl.BBoxCalculatorTest
[INFO] Running org.locationtech.spatial4j.shape.BufferedLineStringTest
Laps: 3325 CWIDbD: 39,110,665,57,2454
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.104 s - in org.locationtech.spatial4j.shape.BufferedLineStringTest
[INFO]
[INFO] Results:
[INFO]
[WARNING] Tests run: 466, Failures: 0, Errors: 0, Skipped: 4
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.868 s
[INFO] Finished at: 2026-05-01T12:41:33+01:00
[INFO] -----
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$
```

Profile

```
projectname='spatial4j'
```

```
java -cp ../gin/build/gin.jar gin.util.Profiler -p $projectname -d ./ -h  
~/sdkman/candidates/maven/current/ -o $projectname.Profiler_output.csv &>  
$projectname.Profiler_stderrstdout.txt
```

```
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ projectname='spatial4j'  
java -cp ../gin/build/gin.jar gin.util.Profiler -p $projectname -d ./ -h ~/sdkman/candidates/maven/current/ -o $projectname.Profiler_output.csv &> $projectname.Profiler_stderrstdout.txt  
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$
```

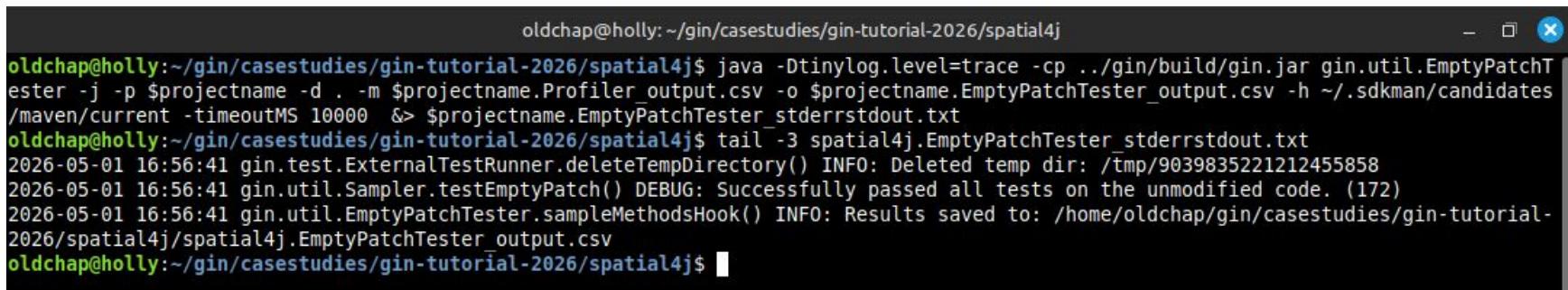
Project	MethodIndex	Method	Count	Tests
spatial4j	1	org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	208	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testPolygonRepair [], org.locationtech
spatial4j	2	org.locationtech.spatial4j.context.SpatialContextFactory.checkDef	134	org.locationtech.spatial4j.io.TestGeohashUtils.testDecodePreciseLongitudeLatitude [], org.I
spatial4j	3	org.locationtech.spatial4j.shape.jts.JtsGeometry.cutUnwrappedG	119	org.locationtech.spatial4j.shape.JtsGeometryTest.testArea [], org.locationtech.spatial4j.sha
spatial4j	4	org.locationtech.spatial4j.context.SpatialContextFactory.makeCla	57	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWriteThenReadBuffe
spatial4j	5	org.locationtech.spatial4j.context.SpatialContextFactory.addRead	56	org.locationtech.spatial4j.io.TestGeohashUtils.testDecodePreciseLongitudeLatitude [], org.I
spatial4j	6	org.locationtech.spatial4j.context.jts.JtsSpatialContextFactory.get	30	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadRectangle [], org.loc
spatial4j	7	org.locationtech.spatial4j.shape.jts.JtsShapeFactory.pointXYZ(dot	25	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint [], org.locationtech.spa
spatial4j	8	org.locationtech.spatial4j.context.jts.JtsSpatialContextFactory.che	21	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParseMultiLineString [], org.location
spatial4j	9	org.locationtech.spatial4j.io.jackson.ShapeDeserializer.read(JsonP	18	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testParseMultiPoint [], c
spatial4j	10	org.locationtech.spatial4j.shape.jts.JtsGeometry.unwrapDateline(0	15	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testPolygonRepair [], org.locationtech
spatial4j	11	org.locationtech.spatial4j.context.SpatialContextFactory.makeFor	14	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadBufferedLine [], org.
spatial4j	12	org.locationtech.spatial4j.shape.jts.JtsGeometry.relate(Geometry)	9	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testPolygonRepair [], org.locationtech
spatial4j	13	org.locationtech.spatial4j.shape.jts.JtsShapeFactory.getGeometr	7	org.locationtech.spatial4j.shape.jts.JtsShapeFactoryTest.testCircleGeometryConversions
spatial4j	14	org.locationtech.spatial4j.shape.jts.JtsShapeFactory.makeShape	6	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testPolygonRepair []
spatial4j	15	org.locationtech.spatial4j.shape.jts.JtsShapeFactory.makeShape(P	6	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadMultiLineString [], or
spatial4j	16	org.locationtech.spatial4j.io.WKTReader.newState(String)	5	org.locationtech.spatial4j.io.JtsWktShapeParserTest.polyToRectCcwRule [], org.locationtec
spatial4j	17	org.locationtech.spatial4j.io.jackson.ShapeAsGeoJSONSerializer.P	4	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWriteThenReadRect
spatial4j	18	org.locationtech.spatial4j.shape.jts.JtsGeometry.computeGeoBBot	4	org.locationtech.spatial4j.shape.JtsGeometryTest.testMakeRect [], org.locationtech.spatia

You might want to repeat this with reboots in between. Gin provides `gin.util.analysis.MergeProfilerFiles` to combine profiler results

Test the Empty Patch

A sanity check does the target project run successfully within Gin's framework without having changed anything?

```
java -Dtinylog.level=trace -cp ../gin/build/gin.jar gin.util.EmptyPatchTester -j -p $projectname -d .  
-m $projectname.Profiler_output.csv -o $projectname.EmptyPatchTester_output.csv  
-h ~/.sdkman/candidates/maven/current -timeoutMS 10000 &> $projectname.EmptyPatchTester_stderrstdout.txt
```



```
oldchap@holly: ~/gin/casestudies/gin-tutorial-2026/spatial4j  
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ java -Dtinylog.level=trace -cp ../gin/build/gin.jar gin.util.EmptyPatchT  
ester -j -p $projectname -d . -m $projectname.Profiler_output.csv -o $projectname.EmptyPatchTester_output.csv -h ~/.sdkman/candidates  
/maven/current -timeoutMS 10000 &> $projectname.EmptyPatchTester_stderrstdout.txt  
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ tail -3 spatial4j.EmptyPatchTester_stderrstdout.txt  
2026-05-01 16:56:41 gin.test.ExternalTestRunner.deleteTempDirectory() INFO: Deleted temp dir: /tmp/9039835221212455858  
2026-05-01 16:56:41 gin.util.Sampler.testEmptyPatch() DEBUG: Successfully passed all tests on the unmodified code. (172)  
2026-05-01 16:56:41 gin.util.EmptyPatchTester.sampleMethodsHook() INFO: Results saved to: /home/oldchap/gin/casestudies/gin-tutorial-  
2026/spatial4j/spatial4j.EmptyPatchTester_output.csv  
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$
```

Test the Empty Patch

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Test	Pass	Fail	Unit Test	RepNumber	PatchValid	PatchCompiled	TestPassed	TestExecutionTime(ns)	TestCPUTime(ns)	TestTimedOut	TestExceptionType	TestExceptionMessage	AssertionExpectedValue	AssertionActualValue	NoOp	EditsValid
1	0	0	1 org.locationtech.spatial4j.io.JtsWktShapeParserTest.testPoly	1	true	true	true	124898011	48575581	false	N/A	N/A	N/A	N/A	true	
2	0	0	2 org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWrite	1	true	true	true	20880643	11251611	false	N/A	N/A	N/A	N/A	true	
3	0	0	3 org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReader	1	true	true	true	144136444	5895515	false	N/A	N/A	N/A	N/A	true	
4	0	0	4 org.locationtech.spatial4j.shape.JtsGeometryTest.testMake	1	true	true	true	54016296	13282413	false	N/A	N/A	N/A	N/A	true	
5	0	0	5 org.locationtech.spatial4j.io.GeneralWktTest.testWriteThen	1	true	true	true	20496029	9768621	false	N/A	N/A	N/A	N/A	true	
6	0	0	6 org.locationtech.spatial4j.io.JtsWktShapeParserTest.testM	1	true	true	true	9269480	3901601	false	N/A	N/A	N/A	N/A	true	
7	0	0	7 org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReader	1	true	true	true	8263394	5319645	false	N/A	N/A	N/A	N/A	true	
8	0	0	8 org.locationtech.spatial4j.io.JtsBinaryCodecTest.testPoly []	1	true	true	true	15180281	7666347	false	N/A	N/A	N/A	N/A	true	
9	0	0	9 org.locationtech.spatial4j.shape.JtsGeometryTest.testFiji []	1	true	true	true	49283835	5916743	false	N/A	N/A	N/A	N/A	true	
10	0	0	10 org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReader	1	true	true	true	26467113	6241741	false	N/A	N/A	N/A	N/A	true	
11	0	0	11 org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReader	1	true	true	true	17665389	5537023	false	N/A	N/A	N/A	N/A	true	
12	0	0	12 org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriter	1	true	true	true	6068885	3054935	false	N/A	N/A	N/A	N/A	true	
13	0	0	13 org.locationtech.spatial4j.shape.JtsShapeFactoryTest.test	1	true	true	true	15517043	15297735	false	N/A	N/A	N/A	N/A	true	
14	0	0	14 org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReader	1	true	true	true	11884828	4344163	false	N/A	N/A	N/A	N/A	true	

Implement prompt template

Please give me \$COUNT\$ different Java implementations of this code snippet:```

\$DESTINATION\$

```

Each implementation should have the exact same functionality. This code belongs to project \$PROJECT\$.

Wrap all code in curly braces, if it is not already. Do not include any method or class declarations. Label all code as java.

# Random search

```
java -Dtinylog.level=trace -cp ../gin/build/gin.jar gin.util.RandomSampler -j
-p $projectname -d . -m $projectname.Profiler_output.csv
-o $projectname.RandomSampler_output.csv -h ~/.sdkman/candidates/maven/current
-timeoutMS 10000 -et gin.edit.llm.LLMReplaceStatement -pn 10 -ptt prompt.txt
-mt llama3.1:8b -mo 300 &> $projectname.RandomSampler_stderrstdout.txt
```

```
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ java -Dtinylog.level=trace -cp ../gin/build/gin.jar gin.util.RandomSampl
er -j -p $projectname -d . -m $projectname.Profiler_output.csv -o $projectname.RandomSampler_ll_output.csv -h ~/.sdkman/candidates/ma
ven/current -timeoutMS 10000 -et gin.edit.llm.LLMReplaceStatement -pn 30 -ptt prompt.txt -mt llama3.1:8b -mo 300 &> $projectname.RS_l
l_stderrstdout
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$
```



# Random search

| PatchIndex | PatchSize | Patch | MethodIndex | TestIndex | UnitTest                                                                    | RepNumber | PatchValid | PatchCompiled | TestPassed | TestExecutionTime(ns) | TestCPUTime(ns) | TestTimedOut | TestExceptionType | TestExceptionMessage | AssertionExp |
|------------|-----------|-------|-------------|-----------|-----------------------------------------------------------------------------|-----------|------------|---------------|------------|-----------------------|-----------------|--------------|-------------------|----------------------|--------------|
| 18         | 1 !!!     | 1     | 1           | 57        | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParseMultiPolygon []   | 1         | true       | true          | true       | 3546884               | 2271018         | false        | N/A               | N/A                  | N/A          |
| 18         | 1 !!!     | 1     | 1           | 58        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testParse    | 1         | true       | true          | true       | 3228907               | 1750321         | false        | N/A               | N/A                  | N/A          |
| 18         | 1 !!!     | 1     | 1           | 59        | org.locationtech.spatial4j.shape.JtsGeometryTest.testRelations []           | 1         | true       | true          | true       | 8525321               | 2216976         | false        | N/A               | N/A                  | N/A          |
| 19         | 1 !!!     | 15    | 1           | 1         | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadMulti     | 1         | true       | true          | true       | 94419015              | 44078817        | false        | N/A               | N/A                  | N/A          |
| 19         | 1 !!!     | 15    | 2           | 2         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod    | 1         | true       | true          | true       | 123680216             | 6961700         | false        | N/A               | N/A                  | N/A          |
| 19         | 1 !!!     | 15    | 3           | 3         | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParsePolygon []        | 1         | true       | true          | true       | 22810259              | 5330274         | false        | N/A               | N/A                  | N/A          |
| 19         | 1 !!!     | 15    | 4           | 4         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite    | 1         | true       | true          | true       | 15827611              | 4515501         | false        | N/A               | N/A                  | N/A          |
| 19         | 1 !!!     | 15    | 5           | 5         | org.locationtech.spatial4j.shape.JtsGeometryTest.testPolyRelatesToCircle [] | 1         | true       | true          | true       | 31358921              | 7682396         | false        | N/A               | N/A                  | N/A          |
| 19         | 1 !!!     | 15    | 6           | 6         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testEncod    | 1         | true       | true          | true       | 8520446               | 4643530         | false        | N/A               | N/A                  | N/A          |
| 20         | 1 !!!     | 32    | 1           | 1         | org.locationtech.spatial4j.context.SpatialContextFactoryTest.testJtsContext | 1         | false      | false         | false      | 0                     | 0               | false        | N/A               | N/A                  | N/A          |
| 21         | 1 !!!     | 16    | 1           | 1         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.polyToRectCcwRule []    | 1         | false      | false         | false      | 0                     | 0               | false        | N/A               | N/A                  | N/A          |
| 21         | 1 !!!     | 16    | 2           | 2         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParseMultiPoint []  | 1         | false      | false         | false      | 0                     | 0               | false        | N/A               | N/A                  | N/A          |
| 21         | 1 !!!     | 16    | 3           | 3         | org.locationtech.spatial4j.io.ShapeFormatTest.testReadAndWriteTheSame []    | 1         | false      | false         | false      | 0                     | 0               | false        | N/A               | N/A                  | N/A          |
| 21         | 1 !!!     | 16    | 4           | 4         | org.locationtech.spatial4j.shape.JtsGeometryTest.testRegressions []         | 1         | false      | false         | false      | 0                     | 0               | false        | N/A               | N/A                  | N/A          |
| 21         | 1 !!!     | 16    | 5           | 5         | org.locationtech.spatial4j.io.WktShapeParserTest.testNoOp []                | 1         | false      | false         | false      | 0                     | 0               | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 1           | 1         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint []       | 1         | true       | true          | true       | 84391513              | 39672742        | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 2           | 2         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint_invalid  | 1         | true       | true          | true       | 8171718               | 4915938         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 3           | 3         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite    | 1         | true       | true          | true       | 131560330             | 6677250         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 4           | 4         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testPoint    | 1         | true       | true          | true       | 16370829              | 4943295         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 5           | 5         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite    | 1         | true       | true          | true       | 9389905               | 5237168         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 6           | 6         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite    | 1         | true       | true          | true       | 8498442               | 5533645         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 7           | 7         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testParse    | 1         | true       | true          | true       | 8256047               | 5311184         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 8           | 8         | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodeBufferedLine     | 1         | true       | true          | true       | 7379638               | 4671208         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 9           | 9         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite    | 1         | true       | true          | true       | 7191912               | 4366098         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 10          | 10        | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodePoint []         | 1         | true       | true          | true       | 6456532               | 4108358         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 11          | 11        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod    | 1         | true       | true          | true       | 8945833               | 4145103         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 12          | 12        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod    | 1         | true       | true          | true       | 5010103               | 2960989         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 13          | 13        | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadLine      | 1         | true       | true          | true       | 5179988               | 3039921         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 14          | 14        | org.locationtech.spatial4j.io.LegacyShapeReadWriterTest.testPoint []        | 1         | true       | true          | true       | 5001429               | 2125105         | false        | N/A               | N/A                  | N/A          |
| 22         | 1 !!!     | 7     | 15          | 15        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite    | 1         | true       | true          | true       | 6738079               | 2876157         | false        | N/A               | N/A                  | N/A          |

# Random search

| PatchIndex | PatchSize | MethodIndex | TestIndex | UnitTest                                                                       | RepNumber                                          |
|------------|-----------|-------------|-----------|--------------------------------------------------------------------------------|----------------------------------------------------|
| 18         | 1 !!!     | 1           | 57        | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParseMultiPolygon []      |                                                    |
| 18         | 1 !!!     | 1           | 58        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testParse       |                                                    |
| 18         | 1 !!!     | 1           | 59        | org.locationtech.spatial4j.shape.JtsGeometryTest.testRelations []              |                                                    |
| 19         | 1 !!!     | 15          | 1         | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadMulti        |                                                    |
| 19         | 1 !!!     | 15          | 2         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod       |                                                    |
| 19         | 1 !!!     | 15          | 3         | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParsePolygon []           |                                                    |
| 19         | 1 !!!     | 15          | 4         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite       |                                                    |
| 19         | 1 !!!     | 15          | 5         | org.locationtech.spatial4j.shape.JtsGeometryTest.testPolyRelatesToCircle []    |                                                    |
| 19         | 1 !!!     | 15          | 6         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testEnco        |                                                    |
| 20         | 1 !!!     | 32          | 1         | org.locationtech.spatial4j.context.SpatialContextFactoryTest.testJtsContext    |                                                    |
| 21         | 1 !!!     | 16          | 1         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testPolyFromGeoJsonRule [] |                                                    |
| 21         | 1 !!!     | 16          | 2         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParseMultiPoint []     |                                                    |
| 21         | 1 !!!     | 16          | 3         | org.locationtech.spatial4j.io.ShapeFormatTest.testReadAndWriteTheSame []       |                                                    |
| 21         | 1 !!!     | 16          | 4         | org.locationtech.spatial4j.shape.JtsGeometryTest.testRegressions []            |                                                    |
| 21         | 1 !!!     | 16          | 5         | org.locationtech.spatial4j.io.WktShapeParserTest.testNoOp []                   |                                                    |
| 22         | 1 !!!     | 7           | 1         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint []          |                                                    |
| 22         | 1 !!!     | 7           | 2         | org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint_invalid     |                                                    |
| 22         | 1 !!!     | 7           | 3         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite       |                                                    |
| 22         | 1 !!!     | 7           | 4         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testPoint       |                                                    |
| 22         | 1 !!!     | 7           | 5         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite       |                                                    |
| 22         | 1 !!!     | 7           | 6         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite       |                                                    |
| 22         | 1 !!!     | 7           | 7         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testParse       |                                                    |
| 22         | 1 !!!     | 7           | 8         | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodeBufferedLine        |                                                    |
| 22         | 1 !!!     | 7           | 9         | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite       |                                                    |
| 22         | 1 !!!     | 7           | 10        | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodePoint []            |                                                    |
| 22         | 1 !!!     | 7           | 11        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod       |                                                    |
| 22         | 1 !!!     | 7           | 12        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod       |                                                    |
| 22         | 1 !!!     | 7           | 13        | org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadLine         | 1 true true true 5179988 3039921 false N/A N/A N/A |
| 22         | 1 !!!     | 7           | 14        | org.locationtech.spatial4j.io.LegacyShapeReadWriterTest.testPoint []           | 1 true true true 5001429 2125105 false N/A N/A N/A |
| 22         | 1 !!!     | 7           | 15        | org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite       | 1 true true true 6738079 2876157 false N/A N/A N/A |

```

| gin.edit.llm.LLMReplaceStatement
"src/main/java/org/locationtech/spatial4j/shape/jts/JtsShapeFactory.java":2215
Prompt: !!!
Please give me 5 different Java implementations of this code snippet:```
{
 JtsGeometry jtsGeom = new JtsGeometry(geom, (JtsSpatialContext) ctx,
dateline180Check, allowMultiOverlap);
 if (isAutoIndex()) {
 jtsGeom.index();
 }
 return jtsGeom;
}
```

```

Each implementation should have the exact same functionality. This code belongs to project spatial4j.

Wrap all code in curly braces, if it is not already. Do not include any method or class declarations. Label all code as java.

!!! --> !!!

```

{
    JtsGeometry jtsGeom = new JtsGeometry(geom, (JtsSpatialContext) ctx,
dateline180Check, allowMultiOverlap);
    if (isAutoIndex()) {
        jtsGeom.index();
    }
    return jtsGeom;
}

!!! |

```

TestTimedOut	TestExceptionType	TestExceptionMessage	AssertionExpectedValue	AssertionActualValue
FALSE	java.lang.AssertionError	expected:<255> but was:<207>	255	207
FALSE	N/A	N/A	N/A	N/A
FALSE	N/A	N/A	N/A	N/A

Random search

PatchIndex	PatchSize	Pat	MethodIndex	TestIndex	UnitTest	RepNumber
18	1 !!!	▶	1	57	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParseMultiPolygon []	
18	1 !!!	▶	1	58	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testParse▶	
18	1 !!!	▶	1	59	org.locationtech.spatial4j.shape.JtsGeometryTest.testRelations []	
19	1 !!!	▶	15	1	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadMulti▶	
19	1 !!!	▶	15	2	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEnco▶	
19	1 !!!	▶	15	3	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParsePolygon []	
19	1 !!!	▶	15	4	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite▶	
19	1 !!!	▶	15	5	org.locationtech.spatial4j.shape.JtsGeometryTest.testPolyRelatesToCircle []	
19	1 !!!	▶	15	6	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testEnco▶	
20	1 !!!	▶	32	1	org.locationtech.spatial4j.context.SpatialContextFactoryTest.testJtsContext▶	
21	1 !!!	▶	16	1	org.locationtech.spatial4j.io.JtsWktShapeParserTest.polyToRectCowRule []	
21	1 !!!	▶	16	2	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParseMultiPoint []	
21	1 !!!	▶	16	3	org.locationtech.spatial4j.io.ShapeFormatTest.testReadAndWriteTheSame []	
21	1 !!!	▶	16	4	org.locationtech.spatial4j.shape.JtsGeometryTest.testRegreptions []	
21	1 !!!	▶	16	5	org.locationtech.spatial4j.io.WktShapeParserTest.testNoOp []	
22	1 !!!	▶	7	1	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint []	
22	1 !!!	▶	7	2	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint_invalid▶	
22	1 !!!	▶	7	3	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWriteT▶	
22	1 !!!	▶	7	4	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testPoint▶	
22	1 !!!	▶	7	5	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite▶	
22	1 !!!	▶	7	6	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWriteT▶	
22	1 !!!	▶	7	7	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testPars▶	
22	1 !!!	▶	7	8	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodeBufferedLine▶	
22	1 !!!	▶	7	9	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWriteT▶	
22	1 !!!	▶	7	10	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodePoint []	
22	1 !!!	▶	7	11	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEnco▶	
22	1 !!!	▶	7	12	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEnco▶	
22	1 !!!	▶	7	13	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadLine▶	1 true true true 5179988 3039921 false N/A N/A N/A
22	1 !!!	▶	7	14	org.locationtech.spatial4j.io.LegacyShapeReadWriterTest.testPoint []	1 true true true 5001429 2125105 false N/A N/A N/A
22	1 !!!	▶	7	15	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite▶	1 true true true 6738079 2876157 false N/A N/A N/A

```
| gin.edit.llm.LLMReplaceStatement
"src/main/java/org/locationtech/spatial4j/context/SpatialContextFactory.java":245
Prompt: !!!
Please give me 5 different Java implementations of this code snippet:```
{
    Class<?> c = classLoader.loadClass(cname);
    instance = (SpatialContextFactory) c.newInstance();
}
...`
```

Each implementation should have the exact same functionality. This code belongs to project spatial4j.

Wrap all code in curly braces, if it is not already. Do not include any method or class declarations. Label all code as java.

!!! --> !!!

LLM GAVE NO SUGGESTIONS

!!! |

TestTimedOut	TestExceptionType	TestExceptionMessage	AssertionExpectedValue	AssertionActualValue
FALSE	java.lang.AssertionError	expected:<255> but was:<207>	255	207
FALSE	N/A	N/A	N/A	N/A
FALSE	N/A	N/A	N/A	N/A

Random search

PatchIndex	PatchSize	Pat	MethodIndex	TestIndex	UnitTest	RepNumber
18	1 !!!	▶	1	57	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParseMultiPolygon []	
18	1 !!!	▶	1	58	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testParse	
18	1 !!!	▶		59	org.locationtech.spatial4j.shape.JtsGeometryTest.testRelations []	
19	1 !!!	▶	15	1	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadMulti	
19	1 !!!	▶	15	2	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod	
19	1 !!!	▶	15	3	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testParsePolygon []	
19	1 !!!	▶	15	4	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite	
19	1 !!!	▶	15	5	org.locationtech.spatial4j.shape.JtsGeometryTest.testPolyRelatesToCircle []	
19	1 !!!	▶	15	6	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testEncod	
20	1 !!!	▶	32	1	org.locationtech.spatial4j.context.SpatialContextFactoryTest.testContext	
21	1 !!!	▶	16	1	org.locationtech.spatial4j.io.JtsWktShapeParserTest.polyToRectCowRule []	
21	1 !!!	▶	16	2	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParseMultiPoint []	
21	1 !!!	▶	16	3	org.locationtech.spatial4j.io.ShapeFormatTest.testReadAndWriteTheSame []	
21	1 !!!	▶	16	4	org.locationtech.spatial4j.shape.JtsGeometryTest.testRegressions []	
21	1 !!!	▶	16	5	org.locationtech.spatial4j.io.WktShapeParserTest.testNoOp []	
22	1 !!!	▶	7	1	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint []	
22	1 !!!	▶	7	2	org.locationtech.spatial4j.io.JtsWktShapeParserTest.testParsePoint_invalid	
22	1 !!!	▶	7	3	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite	
22	1 !!!	▶	7	4	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testPoint	
22	1 !!!	▶	7	5	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite	
22	1 !!!	▶	7	6	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite	
22	1 !!!	▶	7	7	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testParse	
22	1 !!!	▶	7	8	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodeBufferedLine	
22	1 !!!	▶	7	9	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testWrite	
22	1 !!!	▶	7	10	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testEncodePoint []	
22	1 !!!	▶	7	11	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod	
22	1 !!!	▶	7	12	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONWriterTest.testEncod	
22	1 !!!	▶	7	13	org.locationtech.spatial4j.io.GeneralGeoJSONTest.testWriteThenReadLine	1 true true true 5179988 3039921 false N/A N/A N/A
22	1 !!!	▶	7	14	org.locationtech.spatial4j.io.LegacyShapeReadWriterTest.testPoint []	1 true true true 5001429 2125105 false N/A N/A N/A
22	1 !!!	▶	7	15	org.locationtech.spatial4j.io.jackson.JacksonGeoJSONReaderTest.testWrite	1 true true true 6738079 2876157 false N/A N/A N/A

```
| gin.edit.llm.LLMReplaceStatement
"src/main/java/org/locationtech/spatial4j/shape/jts/JtsShapeFactory.java":422
Prompt: !!!
```

Please give me 5 different Java implementations of this code snippet:```\n{\n return ((JtsGeometry) shape).getGeom();\n}\n```\n...

Each implementation should have the exact same functionality. This code belongs to project spatial4j.

Wrap all code in curly braces, if it is not already. Do not include any method or class declarations. Label all code as java.

!!! --> !!!

```
{\n    if (shape instanceof JtsGeometry) {\n        return ((JtsGeometry) shape).getGeom();\n    } else {\n        // Add error handling or throw an exception here\n        return null;\n    }\n}\n!!! |
```

TestTimedOut	TestExceptionType	TestExceptionMessage	AssertionExpectedValue	AssertionActualValue
FALSE	java.lang.AssertionError	expected:<255> but was:<207>	255	207
FALSE	N/A	N/A	N/A	N/A
FALSE	N/A	N/A	N/A	N/A

Local Search

```
java -Dtinylog.level=trace -cp ../gin/build/gin.jar gin.util.LocalSearchRuntime -j  
-p $projectname -d . -m $projectname.Profiler_output.csv  
-o $projectname.LocalSearchRuntime_output.csv -h ~/.sdkman/candidates/maven/current  
-timeoutMS 10000 -et gin.edit.llm.LLMReplaceStatement -ptt prompt.txt  
-mt llama3.1:8b -in 200 &> $projectname.LocalSearchRuntime_LLM_10_stderrstdout.txt
```

```
oldchap@holly:~/gin/casestudies/gin-tutorial-2026/spatial4j$ java -Dtinylog.level=trace -cp ../gin/build/gin.jar gin.util.LocalSearch  
Runtime -j -p $projectname -d . -m $projectname.Profiler_output.csv -o $projectname.LocalSearchRuntime_output.csv -h ~/.sdkman/candid  
ates/maven/current -timeoutMS 10000 -et gin.edit.llm.LLMReplaceStatement -ptt prompt.txt -mt llama3.1:8b -in 200 &> $projectname.Loca  
lSearchRuntime_LLM_10_stderrstdout.txt
```

This will run a hillclimber for each “hot” method listed in the profiler output

Local Search

186	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	184	184 !!!	true	true	845.6264	845	241
187	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	185	185 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
188	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	186	186 !!!	true	true	870.64764	870	216
189	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	187	187 !!!	true	true	871.0221	871	215
190	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	188	188 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
191	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	189	189 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
192	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	190	190 !!!	true	true	846.6334	846	240
193	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	191	191 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
194	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	192	192 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
195	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	193	193 !!!	true	true	877.468	877	209
196	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	194	194 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
197	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	195	195 !!!	true	true	1026.7664	1026	60
198	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	196	196 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
199	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	197	197 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
200	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	198	198 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
201	org.locationtech.spatial4j.shape.jts.JtsGeometry.org.locationtech.spatial4j.shape.jts.JtsGeometry.validate()	199	199 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
202	org.locationtech.spatial4j.context.SpatialContextFactory.org.locationtech.spatial4j.context.SpatialContextFa	-1	0	true	true	1255.0575	1255	0
203	org.locationtech.spatial4j.context.SpatialContextFactory.org.locationtech.spatial4j.context.SpatialContextFa	1	1 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
204	org.locationtech.spatial4j.context.SpatialContextFactory.org.locationtech.spatial4j.context.SpatialContextFa	2	2 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308
205	org.locationtech.spatial4j.context.SpatialContextFactory.org.locationtech.spatial4j.context.SpatialContextFa	3	3 !!!	false	false	0	1.7976931348623157E+308	-1.7976931348623157E+308

Local Search

```
2026-05-03 09:24:16 gin.test.ExternalTestRunner.runTests() DEBUG: Running test 19/20 in module '': rep=1/1, testIndex=19/20:
```

```
org.locationtech.spatial4j.shape.JtsGeometryTest.testRelations []
```

```
2026-05-03 09:24:17 gin.test.ExternalTestRunner.deleteTempDirectory() INFO: Deleted temp dir: /tmp/16740013997709357880
```

```
2026-05-03 09:24:17 gin.util.LocalSearchSimple.search() INFO: New best patch found: | gin.edit.llm.LLMReplaceStatement
```

```
"src/main/java/org/locationtech/spatial4j/context/jts/JtsSpatialContextFactory.java":240
```

```
Prompt: !!!
```

```
Please give me 5 different Java implementations of this code snippet:``
```

```
{
  writers.add(JtsGeoJSONWriter.class);
  writers.add(JtsWKTWriter.class);
  writers.add(JtsPolyshapeWriter.class);
  writers.add(LegacyShapeWriter.class);
}
...

```

Each implementation should have the exact same functionality. This code belongs to project spatial4j.

Wrap all code in curly braces, if it is not already. Do not include any method or class declarations. Label all code as java.

```
!!! --> !!!
```

```
{
  writers.add(JtsGeoJSONWriter.class);
}

```

```
!!! | gin.edit.llm.LLMReplaceStatement "src/main/java/org/locationtech/spatial4j/context/jts/JtsSpatialContextFactory.java":240
```

```
Prompt: !!!
```

```
NOT YET APPLIED
```

```
!!! --> !!!
```

```
NOT YET APPLIED
```

```
!!! | with fitness: 410.0
```

```
2026-05-03 09:24:17 gin.Patch.randomEdit() INFO: Choosing an edit type from: [class gin.edit.llm.LLMReplaceStatement]
```

Results

Can easily extract from CSVs:

Valid/Compile/Test pass rates

Run time

Specific exceptions

PatchAnalyser and PatchExtractor tools are provided to investigate patched code...

Results

```
java -cp ../gin/build/gin.jar gin.PatchExtractor  
-pl spatial4j.LocalSearchRuntime_output.csv
```

The screenshot displays two windows. The left window, titled 'spatial4j.LocalSearchRuntime_output.csv_patch_3', shows a file explorer view with a table of files:

Name	Size	Type	Date Modified
JtsGeometry.blocks	35.8 kB	Text	Mon 04 May 2026 12:14:32 BST
JtsGeometry.original	28.1 kB	Text	Mon 04 May 2026 12:14:32 BST
JtsGeometry.patched	28.2 kB	Text	Mon 04 May 2026 12:14:32 BST
JtsGeometry.statements	75.4 kB	Text	Mon 04 May 2026 12:14:32 BST

The right window, titled 'Meld', shows a diff between 'JtsGeometry.original' and 'JtsGeometry.patched'. The diff highlights changes in the `validate()` method. The original code (left) has a try-catch block for `InvalidShapeException`. The patched code (right) adds a `try` block around the `isValidOp.isValid()` check, which is highlighted in blue. The catch block is highlighted in green.

```
133  */  
134  public void validate() throws InvalidShape-  
Exception {  
135      if (!validated) {  
136          IsValidOp isValidOp = new IsValidOp  
(geom);  
137      if (!isValidOp.isValid())  
138          throw new InvalidShapeException  
(isValidOp.getValidationError().toString());  
139      validated = true;  
140  }  
141  }  
142  /**  
143   * Determines if the shape has been indexed  
144   */  
145  boolean isIndexed() {  
146      ...  
147  }  
148  }
```

The diff shows the following changes in the patched version:

- Line 137: Added `try {` before the `if (!isValidOp.isValid())` check.
- Line 138: The `if (!isValidOp.isValid())` check is now inside the try block.
- Line 139: Added `return;` after the `if` statement.
- Line 140: The `throw new InvalidShapeException` statement is now outside the try block.
- Line 141: Added `} catch (InvalidShapeException e) {` to the end of the try block.
- Line 142: Added `}` to close the catch block.

Gin papers

Available from <https://github.com/gintool/gin>

Cite these:

Gin v2: GECCO 2019

Gin: Genetic Improvement Research Made Easy

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Gin LLM support

Automated Software Engineering (2025) 32:15
<https://doi.org/10.1007/s10515-024-00473-6>

Large language model based mutations in genetic improvement

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Alina Geiger⁴ · Carol Hanna² · Justyna Petke² · Federica Sarro² ·
Dominik Sobania⁴

MAGPIE

Machine Automated General Performance Improvement via
Evolution of software

<https://github.com/bloa/magpie>

Developed by Aymeric Blot

Based on PyGGI

arXiv:2208.02811v1

MAGPIE: MACHINE AUTOMATED GENERAL PERFORMANCE
IMPROVEMENT VIA EVOLUTION OF SOFTWARE

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MAGPIE

Language-agnostic (has examples in c, c++, java, python, ruby)

Supports line and tree level edits

Can target functional and non-functional objectives (uses std output)

Implemented in Python (3.11), configured through scenarios

http://www0.cs.ucl.ac.uk/staff/a.blot/files/slides/blot_gi@icse_2025.pdf

<https://youtu.be/V67elsoKZNM>

MAGPIE

Also designed to
be hacked

Made of simple
building blocks

```
class LineSwapEdit(AbstractEdit):
    @classmethod
    def auto_create(cls, ref):
        target = ref.random_model(AbstractLineModel, writable=True).random_target('line')
        ingredient = ref.random_model(AbstractLineModel, writable=True).random_target('line')
        if not (target and ingredient):
            return None
        return cls(target, ingredient)

    def apply(self, ref, variant):
        ingredient = self.data[0]
        ref_model = ref.models[ingredient[0]]
        model = variant.models[self.target[0]]
        b1 = model.do_replace(ref_model, self.target, ingredient)
        b2 = model.do_replace(ref_model, ingredient, self.target)
        return b1 or b2
```

```
import enum
import time

class TriangleType(enum.Enum):
    INVALID, EQUILATERAL, ISOSCELES, SCALENE = 0, 1, 2, 3

def delay():
    time.sleep(0.001)

def classify_triangle(a, b, c):
    # slow down execution
    delay()

    # sort the sides so that a <= b <= c
    if a > b:
        tmp = a
        a = b
        b = tmp

    if a > c:
        tmp = a
        a = c
        c = tmp

    if b > c:
        tmp = b
        b = c
        c = tmp

    if a + b <= c:
        return TriangleType.INVALID
    elif a == b and b == c:
        return TriangleType.EQUILATERAL
    elif a == b or b == c:
        return TriangleType.ISOSCELES
    else:
        return TriangleType.SCALENE
```

MAGPIE Example

target

```
import time
from triangle import classify_triangle

# examples taken from test_triangle.py
triangles = [
    # invalid
    (1, 2, 9), (1, 9, 2), (2, 1, 9), (2, 9, 1), (9, 1, 2), (9, 2, 1),
    (1, 1, -1), (1, -1, 1), (-1, 1, 1),
    # equilateral
    (1, 1, 1), (100, 100, 100), (99, 99, 99),
    # isosceles
    (100, 90, 90), (90, 100, 90), (90, 90, 100), (2, 2, 3),
    # scalene
    (5, 4, 3), (5, 3, 4), (4, 5, 3), (4, 3, 5), (3, 5, 4),
]

if __name__ == '__main__':
    # we don't really care about the output (we know the function work, from the test suite)
    # we just want to have something on which to measure running time

    for _ in range(3):
        for triangle in triangles:
            classify_triangle(*triangle)
```

MAGPIE Example runner

```
import pytest
from triangle import TriangleType, classify_triangle
```

```
@pytest.mark.parametrize('triangle', [
    (1, 2, 9),
    (1, 9, 2),
    (2, 1, 9),
    (2, 9, 1),
    (9, 1, 2),
    (9, 2, 1),
    (1, 1, -1),
    (1, -1, 1),
    (-1, 1, 1),
])
def test_invalid_triangles(triangle):
    assert classify_triangle(*triangle) == TriangleType.INVALID

@pytest.mark.parametrize('triangle', [
    (1, 1, 1),
    (99, 99, 99),
    (100, 100, 100),
])
def test_equilateral_triangles(triangle):
    assert classify_triangle(*triangle) == TriangleType.EQUILATERAL

@pytest.mark.parametrize('triangle', [
    (100, 90, 90),
    (90, 100, 90),
    (90, 90, 100),
    (2, 2, 3),
])
def test_isosceles_triangles(triangle):
    assert classify_triangle(*triangle) == TriangleType.ISOSCELES
...

```

MAGPIE Example

tests

Runtime example

```
[srcml]
```

```
focus = stmt block
```

```
internodes = block
```

```
[software]
```

```
path = examples/triangle-py
```

```
target_files =
```

```
    triangle.py.xml
```

```
fitness = time
```

```
init_cmd = bash init_slow.sh
```

```
test_cmd = pytest
```

```
run_cmd = python run_triangle.py
```

```
run_timeout = 1
```

```
[search]
```

```
max_steps = 100
```

```
max_time = 60
```

```
possible_edits =
```

```
    XmlNodeDeletion<stmt>
```

```
    XmlNodeReplacement<stmt>
```

```
    XmlNodeInsertion<stmt,block>
```

Runtime example

```
[srcml]  
focus = stmt block  
internodes = block
```

```
init_cmd = bash init_slow.sh  
test_cmd = pytest  
run_cmd = python run_triangle.py  
run_timeout = 1
```

```
#!/usr/bin/env bash  
  
python3 ../../../../magpie python_to_xml triangle.py > triangle.py.xml
```

```
triangle.py.xml  
fitness = time
```

```
max_time = 60  
possible_edits =  
    XmlNodeDeletion<stmt>  
    XmlNodeReplacement<stmt>  
    XmlNodeInsertion<stmt,block>
```

```
oldchap@holly: ~/magpie/magpie
(magpie311) oldchap@holly:~/magpie/magpie$ python magpie local_search --scenario examples/triangle-py/_magpie/scenario_slow.txt
==== SEARCH: FirstImprovement ====
~~~~ WARMUP ~~~~
WARM    SUCCESS          0.11 (--) [0 edit(s)]
WARM    SUCCESS          0.11 (--) [0 edit(s)]
WARM    SUCCESS          0.11 (--) [0 edit(s)]
REF     SUCCESS          0.11 (--) [0 edit(s)]

~~~~ START ~~~~
1  TEST_LENGTHOUT      None (--) [1 edit(s)]
2  TEST_LENGTHOUT      None (--) [1 edit(s)]
3  SUCCESS              *0.10 (94.72%) [1 edit(s)]
4  TEST_CODE_ERROR     None (--) [1 edit(s)]
```

```

97 SUCCESS 0.07 (51.96%) [7 edit(s)]
98 SUCCESS 0.06 (42.72%) [5 edit(s)]
99 SUCCESS +0.04 (27.48%) [5 edit(s)] [cached]
100 SUCCESS 0.07 (51.46%) [6 edit(s)]
~~~ END ~~~

==== REPORT ====
Termination: step budget
Log file: /home/oldchap/magpie/magpie/_magpie_logs/triangle-py_20260503_1777826142.log
Patch file: _magpie_logs/triangle-py_20260503_1777826142.patch
Diff file: _magpie_logs/triangle-py_20260503_1777826142.diff
Reference fitness: 0.13
Best fitness: 0.04

==== BEST PATCH ====
XmlNodeInsertion<stmt,block>(('triangle.py.xml', 'inter block', 6), ('triangle.py.xml', 'stmt', 2)) | XmlNodeInsertion<stmt,block>(('triangle.py.xml', 'inter block', 3), ('triangle.py.xml', 'stmt', 2)) | XmlNodeReplacement<stmt>(('triangle.py.xml', 'stmt', 7), ('triangle.py.xml', 'stmt', 12)) | XmlNodeReplacement<stmt>(('triangle.py.xml', 'stmt', 8), ('triangle.py.xml', 'stmt', 3))

==== DIFF ====
--- before: triangle.py
+++ after: triangle.py
@@ -2,6 +2,10 @@
import enum
import time

+class TriangleType(enum.Enum):
+    class TriangleType(enum.Enum):
+        INVALID, EQUILATERAL, ISOSCELES, SCALENE = (0, 1, 2, 3)
+        INVALID, EQUILATERAL, ISOSCELES, SCALENE = (0, 1, 2, 3)
+class TriangleType(enum.Enum):
+    INVALID, EQUILATERAL, ISOSCELES, SCALENE = (0, 1, 2, 3)

@@ -9,11 +13,11 @@
time.sleep(0.001)

def classify_triangle(a, b, c):
-    delay()
-    if a > b:
+    if a > c:
+        tmp = a
+        a = b
+        b = tmp
+    a = c
+    c = tmp
+    INVALID, EQUILATERAL, ISOSCELES, SCALENE = (0, 1, 2, 3)
    if a > c:
        tmp = a
        a = c

```

```
(magpie311) oldchap@holly:~/magpie/magpie$
```

Overview

- | | | |
|---|--------|----------|
| ● Introduction: why GI? And basic principles | 25 min | [John] |
| ● Recent research questions | 15 min | [Markus] |
| ● The human perspective | 15 min | [John] |
| ● Noteworthy papers & connections to other topics | 15 min | [Markus] |
| ● Demonstration: Gin and Magpie | 30 min | [Sandy] |
| ● Open Challenges and Q&A | 10 min | [Markus] |

Open challenges (1 of 4) – *Trustworthy Fixes*

From “patch found” to “problem solved”

- Many generated patches pass the test suite, but are still wrong. This is often called overfitting: the patch fits the tests, not the real problem.
- The same issue appears in GI more broadly: a change may improve one measured metric, but hurt the software in real use.
- LLM-based repair makes this more urgent, because the generated code can look very convincing even when it is incorrect.

⇒ The field needs stronger ways to check correctness, beyond just running tests.

Open challenges (2 of 4) –

Better Search Spaces

We still search in software spaces that are hard to search well

- Many GI and APR methods still work with low-level code edits.
- But software was not represented for search — it was represented for programming and compilation.
- This means many nearby edits are invalid, unhelpful, or hard to interpret.
- As a result, search can waste effort on bad mutations and weak candidates.

⇒ A big research goal is to move from syntax-based edits to semantics-aware edits.

Open challenges (3 of 4) – ***Benchmarks and Realism***

The evaluation problem is now a research problem

- It is still hard to compare APR and GI systems fairly.
- Benchmarks differ in setup, assumptions, and reporting.
- For modern ML and LLM approaches, data leakage is also a risk.
- Many current benchmarks still focus on small, local bugs, while real fixes often need changes across files, tests, and APIs.

⇒ The field needs better benchmarks, clearer evaluation rules, and more repository-scale tasks.

Open challenges (4 of 4) – ***Adoption, Control, and Cost***

Good research results are not enough for real use

- Even strong repair systems will not be adopted unless developers trust them. Developers need explanations, control, and tools that fit their workflow.
- LLM-based repair brings extra cost and latency.
- So the challenge is not only quality — it is also trust, usability, and efficiency.

⇒ The likely future is not fully automatic repair everywhere, but human-centered tools that assist developers in practical ways.

Open challenges – **Outlook**

What will matter most in the next few years?

- Better ways to check that fixes are truly correct.
- Better search spaces and more semantic edit operators.
- Better benchmarks and more realistic evaluation.
- Better integration with developers and engineering workflows.

In short:

*The next phase is not just about generating more patches —
it is about generating reliable, useful, and deployable improvements.*

Questions?

- Introduction: why GI? And basic principles
- Challenges and open research questions
- Case study: fixing bugs
- The human perspective
- Noteworthy papers, and connections to other topics
- Demonstration: Gin
- Open Challenges and Q&A

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